CHAPTER 6

AORTIC VALVE DISORDERS

Michael Ragosta

Clinical decisions in patients with aortic valve disease depend on an accurate estimation of the severity of the valve lesion and an understanding of the spectrum of hemodynamic derangements associated with the disorder. Echocardiography serves as the primary imaging modality for this purpose. In the current era, cardiac catheterization is generally employed as an adjunctive method to answer specific questions in patients with aortic regurgitation and aortic stenosis, such as disease severity if this is unclear on noninvasive testing or if there is discordance between symptoms and disease severity determined by echocardiography. In addition, with the advent of transcatheter aortic valve replacement (TAVR), invasive hemodynamic assessment of aortic valve disease has achieved greater importance in the preprocedural and postprocedural phases.

The cardiologist needs to understand the numerous potential sources of error when assessing the aortic valve. The adage “Bad data are worse than no data at all” is particularly relevant for aortic stenosis, in which the errors in estimating the stenosis severity may lead to an entirely wrong conclusion regarding the need for valve replacement. This chapter reviews the hemodynamic features of aortic regurgitation and aortic stenosis, potential errors in data collection and interpretation, and the hemodynamics observed with transcatheter valve interventions.

AORTIC VALVE REGURGITATION

Regurgitation of the aortic valve may be caused by a variety of conditions (Box 6.1). Currently, in the United States, aortic regurgitation is most commonly due to aortoannular ectasia from hypertension or from a congenitally bicuspid aortic valve. Endocarditis is another frequently seen cause, and structural deterioration of an aged bioprosthetic aortic valve is also common. A mechanical aortic valve may result in regurgitation if thrombosis of the prosthesis occurs and results in a leaflet stuck in an open position.

The pathophysiology and hemodynamic abnormalities observed in aortic regurgitation depend on several variables, including the severity of the regurgitation, whether regurgitation is acute or chronic, and the compensatory response to volume overload intrinsic to this lesion.

Chronic Versus Acute Aortic Regurgitation

Hemodynamic consequences of severe aortic insufficiency depend on whether regurgitation is acute or chronic. In chronic aortic regurgitation, the gradual onset of progressive valve regurgitation leads to compensatory changes, allowing a prolonged state of adaptation and a clinically asymptomatic state. Progressive left ventricular dilatation ensues with stroke volume increasing to maintain forward flow. Both end-diastolic and end-systolic volumes increase, maintaining ejection fraction. With enlargement of the left ventricular chamber, ventricular wall thickness must increase to maintain normal wall stress, as dictated by the law of Laplace, which states that the wall stress is proportional to the product of transmural pressure and radius divided by wall thickness. Traditionally, chronic aortic regurgitation has been considered an example of pure chronic volume overload. However, systolic pressure also rises in association with the augmented stroke volume; thus in chronic severe aortic regurgitation, the left ventricle is both volume and pressure overloaded with compensation consisting of both ventricular dilatation and hypertrophy.[1,2] Chronic aortic regurgitation exceeds other pathological conditions that cause ventricular enlargement and results in the biggest hearts observed clinically (termed *cor bovinum*, Latin for cow heart). Eventually, however, this compensation is exhausted and contractility diminishes, ejection fraction falls, and decompensation occurs, leading to symptoms of heart failure.

The physiologic benefits of vasodilators in chronic severe aortic regurgitation relate primarily to improved left ventricular function and diminished afterload, especially when systolic hypertension is present. Vasodilator agents do not reduce the regurgitant volume, unless associated diastolic hypertension is present, because the amount of regurgitation is based on both the regurgitant orifice area (unaffected by vasodilators) and the mean gradient between the aorta and the left ventricle during diastole. Because aortic diastolic pressures are already low, vasodilators cannot diminish this gradient further without compromising coronary blood flow.

Many of the interesting physical examination findings of chronic severe aortic regurgitation parallel the hemodynamic findings and are consequences of the compensatory mechanisms that reflect primarily the increased left ventricular size and stroke volume. These include a wide arterial pulse pressure that may exceed 100 mm Hg, a carotid “shudder” from the increased stroke volume, the presence of a diffuse and hyperdynamic apical impulse with lateral displacement, and numerous named signs for various peripheral manifestations (Table 6.1). Auscultation of the heart sounds reveals a soft or absent aortic component of the second heart sound and a characteristic decrescendo diastolic murmur. In chronic compensated aortic regurgitation, the severity of regurgitation correlates with the duration rather than the intensity of the murmur; the murmur is holodiastolic in severe aortic regurgitation and is heard only in early diastole with mild aortic regurgitation. A systolic ejection murmur reflects the increased stroke volume. In some cases

Box 6.1 Causes of Aortic Valve Regurgitation

1. Dilatation of the ascending aorta (aortoannular ectasia)
2. Prolapse or incomplete closure of a congenitally bicuspid valve
3. Endocarditis
4. Aortic prosthetic valve failure
5. Rheumatic valvular disease
6. Ankylosing spondylitis
7. Rheumatoid arthritis
8. Ehlers-Danlos syndrome
9. Marfan disease
10. Syphilis
11. Aortic arch aneurysm
12. Aortic dissection
13. Ventricular septal defect (prolapsing cusp)
14. Subaortic membrane

Table 6.1 Peripheral Manifestations of Chronic Severe Aortic Valve Regurgitation

FEATURE	FINDING
Hill sign	Systolic pressure in the popliteal artery exceeds the pressure in the brachial artery
Corrigan pulse de Musset sign	"Water hammer" or collapsing pulse; visible arterial bounding pulse with quick upstroke in the carotids
Quincke sign	Head bobbing
Müller sign	Visible capillary pulsations of the base of the nail beds
Traube sign	Pulsations of the uvula
Duroziez sign	Loud systolic sounds ("pistol shot") over the femoral arteries
	Systolic murmur heard over the femorals with proximal compression of the artery by a stethoscope; diastolic murmur when compressed distally

of chronic severe aortic regurgitation, the regurgitant stream strikes the mitral valve, and the elevations of the left ventricular diastolic pressure may cause early and partial closure of the mitral valve, resulting in a *functionally stenotic* mitral valve with an associated diastolic rumble (the Austin Flint murmur).

Acute aortic regurgitation behaves entirely different. With the development of acute severe aortic regurgition, none of the compensatory mechanisms involved in the adaptation of chronic severe regurgitation, such as progressive left ventricular dilatation and hypertrophy, are possible. Instead, severe sudden onset of aortic regurgitation causes a dramatic rise in the left ventricular diastolic pressure. The inability to augment the stroke volume because of a normal ventricular cavity size causes a profound and life-threatening decrease in forward flow. The only possible compensatory mechanism in the setting of acute severe aortic regurgitation is tachycardia.

Patients with acute aortic regurgitation are critically ill with respiratory failure from pulmonary edema, hypotension, and shock from diminished cardiac output and a compensatory tachycardia. Interestingly, the diagnosis of acute aortic regurgitation may be difficult because the diastolic murmur is typically early and soft in acute aortic insufficiency and may be completely obscured by extensive pulmonary rales from the associated pulmonary edema. The absence of the compensatory mechanisms described previously prevents the development of the classic peripheral signs of chronic severe aortic regurgitation (i.e., wide pulse pressure, Duroziez sign). For these reasons, prompt diagnosis depends on heightened clinical suspicion for the condition; acute aortic regurgitation should always be considered by the clinician faced with an acutely ill patient with pulmonary edema and hypotension of unclear etiology.

Hemodynamic Findings

In chronic well-compensated aortic regurgitation, the hemodynamic findings typically reflect the associated marked increases in the left ventricular volume and stroke volume. The central aortic pressure waveform is characterized by high systolic pressure, low diastolic pressure, and, consequently, a wide pulse pressure (Fig. 6.1). The marked increase

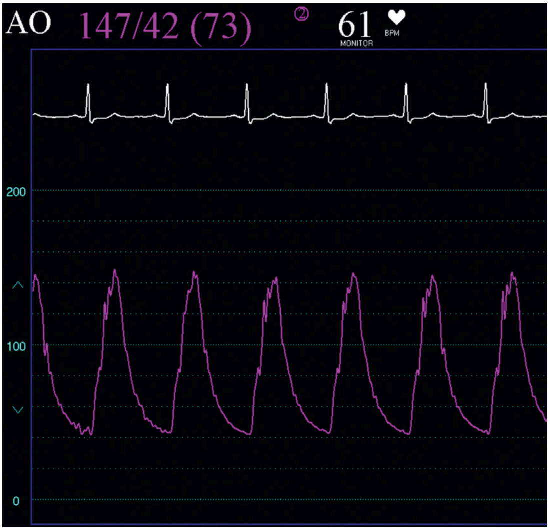


Fig. 6.1 Example of wide pulse pressure seen in a patient with severe chronic aortic (AO) regurgitation. Note the systolic hypertension, a common feature of chronic aortic regurgitation. *BPM*, Beats per minute.

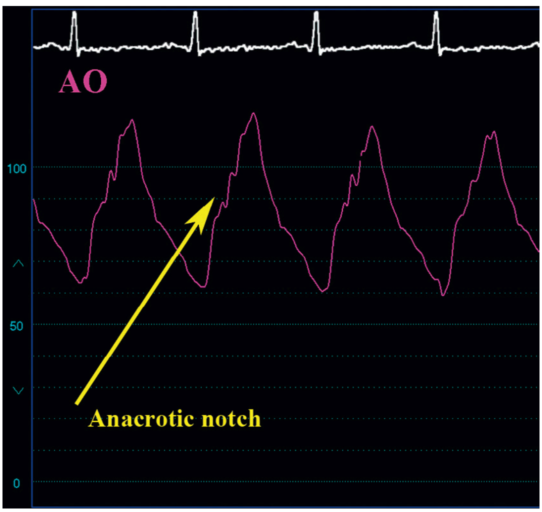


Fig. 6.2 The anacrotic notch, or shoulder *(arrow)*, may appear prominently in cases of severe aortic (AO) regurgitation because of turbulence associated with the abnormal valve. Mild stenosis of this bicuspid valve was also present.

in the stroke volume forms the basis of several hemodynamic findings. Increased systolic flow across an abnormal valve leads to turbulence and a prominent anacrotic notch during systole on the central aortic waveform (Fig. 6.2). The normal physiologic phenomenon of peripheral amplification is even further exaggerated, yielding systolic pressures in the femoral artery greatly exceeding the central aortic pressures (Fig. 6.3). This phenomenon accounts for many of the peripheral signs of aortic regurgitation described earlier.

Additional hemodynamic findings reflect the left ventricular function and the state of compensation. Asymptomatic patients with chronic severe aortic regurgitation whose ventricular function remains normal and who have achieved excellent compensation often exhibit normal or mildly abnormal, resting hemodynamics, except for a wide aortic pulse pressure and marked peripheral amplification. Left ventricular end-diastolic pressure (LVEDP) remains low with

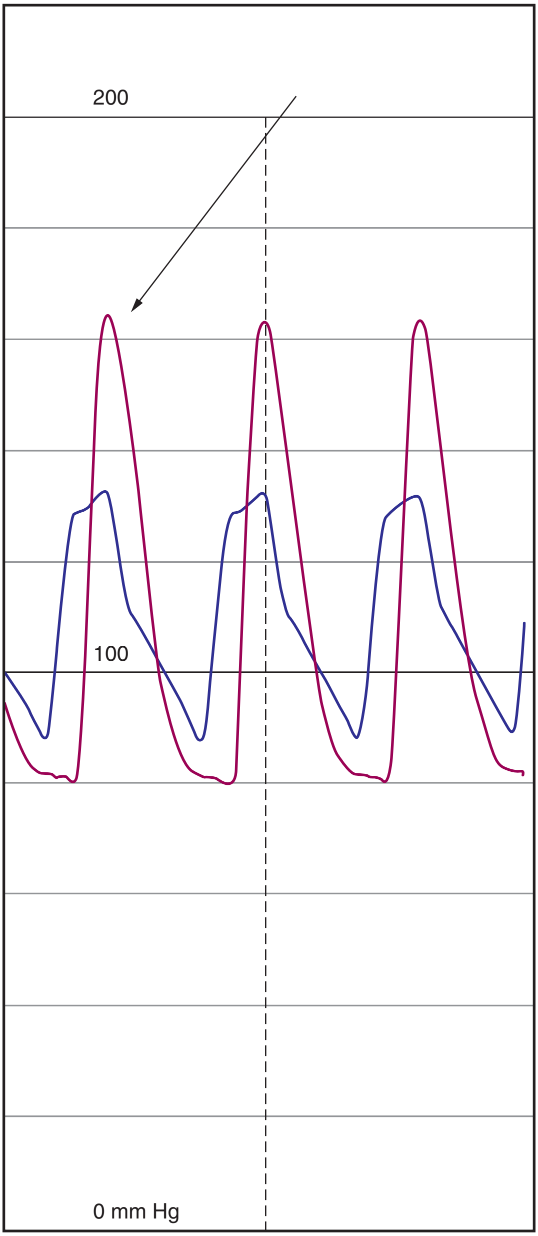


Fig. 6.3 Marked peripheral amplification in the femoral artery pressure *(arrow)* compared with the central aortic pressure in a patient with severe chronic aortic regurgitation.

right-sided pressures unaffected. With the development of symptoms of left ventricular dysfunction, the left ventricular diastolic pressure rises. Typically, early diastolic pressure is normal and then rapidly rises by end diastole. The aortic and ventricular diastolic pressures may become equal in late diastole, a phenomenon known as *diastasis* (Fig. 6.4). The point at which diastasis occurs defines the end of the diastolic murmur; at this phase of the cardiac cycle, flow no longer occurs between the aorta and the left ventricle. This fact accounts for the shortening of the murmur with

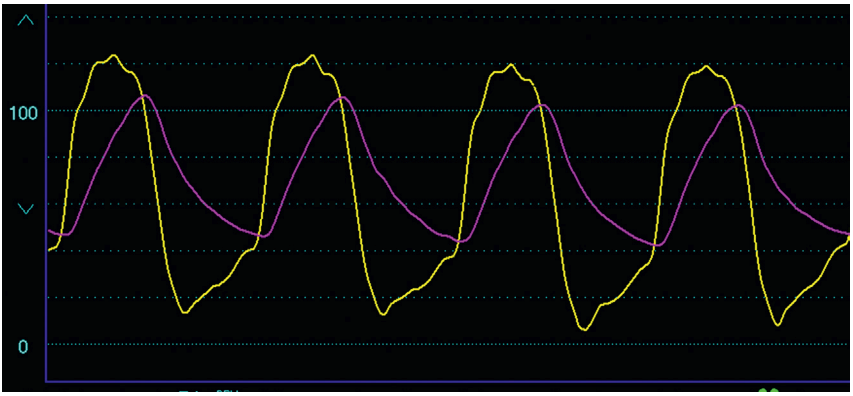


Fig. 6.4 Simultaneous left ventricular and central aortic pressures in a patient with severe aortic regurgitation. Note the rapid rise in the left ventricular diastolic pressure and near equalization of the left ventricular end-diastolic and arterial diastolic pressures *(diastasis)*. A systolic pressure gradient is also noted in this patient with mixed stenosis and regurgitation.

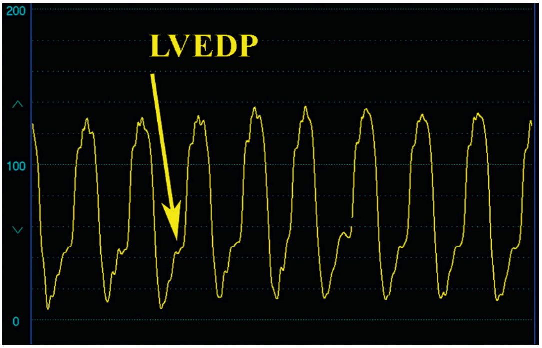


Fig. 6.5 Marked elevation in the left ventricular end-diastolic pressure (LVEDP) *(arrow)* in a patient with severe chronic aortic regurgitation with recent decompensation.

decompensation and the associated rise in the left ventricular diastolic pressure. The degree of elevation of the LVEDP with decompensation varies widely; values more than 50 mm Hg are not unheard of (Fig. 6.5). With the development of decompensation, chronic aortic regurgitation may lead to elevations in right-sided pressures typically associated with heart failure.

Chronic severe aortic regurgitation may elevate the right ventricular end-diastolic pressure in the absence of elevation of the pulmonary capillary wedge pressure or the pulmonary artery systolic pressure. This has been explained on the basis of the *Bernheim* effect, in which the increased left ventricular volume and the elevations in left ventricular diastolic pressures are transmitted to the right ventricle, thus elevating the right ventricular pressures.

With chronic aortic regurgitation, the low aortic diastolic pressure may adversely affect coronary perfusion. The combination of decreased coronary perfusion coupled with increased demand from greater myocardial mass may lead to ischemia from supply-versus-demand mismatch. Importantly, other causes of a wide pulse pressure exist besides chronic severe aortic regurgitation (Fig. 6.6). They include marked bradycardia, severe systolic hypertension, the presence of a rigid and inelastic aorta as often seen in the elderly or in patients with vascular disease, and the presence of a high-output state as observed in patients with severe anemia, hyperthyroidism, anxiety, significant arteriovenous fistulas, or a large patent ductus.

The hemodynamic findings of acute aortic regurgitation reflect the physiologic effects of acute volume overload and diminished forward flow. The left ventricular diastolic pressure increases with a rapidly rising slope, obscuring the *a* wave and culminating in marked elevation of the LVEDP. Diastasis is commonly seen. By mid or late diastole, the left ventricular diastolic pressure may exceed the left atrial pressure, resulting in early closure of the mitral valve (Fig. 6.7).

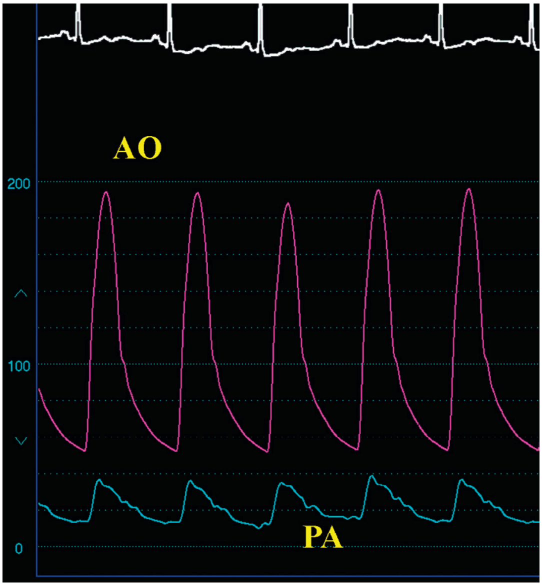


Fig. 6.6 A wide pulse pressure may be present on the aortic (AO) pressure tracing in conditions other than AO insufficiency. This tracing shows a wide pulse pressure in an elderly female with hypertensive heart disease and no AO regurgitation. *PA*, Pulmonary artery.

Pulmonary edema and subsequent elevations in right-sided pressures ensue with associated hypotension and low cardiac output from diminished forward flow, despite a compensatory tachycardia.

An unusual but interesting finding may be present on the aortic pressure trace in acute aortic regurgitation. In the setting of acute severe aortic regurgitation, premature diastolic opening of the aortic valve may occur from marked elevations in the left ventricular diastolic pressure. With a prematurely opened aortic valve, the additional increase in the left ventricular pressure from atrial systole may transmit to the aortic pressure wave, inscribing an *a* wave on the aortic waveform (Fig. 6.8). This rare finding is highly sensitive for acute aortic regurgitation.[3]

Angiography in Aortic Regurgitation

Grading the severity of aortic regurgitation in the cardiac catheterization laboratory is based on angiography and not hemodynamics. The hemodynamic abnormalities described provide information regarding the physiologic consequences of aortic regurgitation and the extent of compensation. Nevertheless, when performing cardiac catheterization on patients with aortic regurgitation, both angiography and hemodynamics are important and provide valuable complementary information to the clinician regarding this valvular lesion. Echocardiographic methods of grading aortic regurgitation have been extensively discussed elsewhere.[4]

Grading the severity of aortic regurgitation is based on contrast aortography of the ascending aorta, using a pigtail catheter positioned carefully above the aortic valve. Optimal opacification usually requires about 60 mL of iodinated contrast delivered rapidly (flow rate of 30 mL/s). A commonly used semiquantitative scale for grading aortic regurgitation is shown in Box 6.2. The degree of regurgitation can also be quantitated by determining the regurgitant volume, which is simply a comparison of the angiographically determined stroke volume with the forward stroke volume determined by the Fick or thermodilution method, as shown by the formula:

$$\text{Regurgitant fraction} = \left(\text{Stroke volume}_{\text{angiography}}\right) - \frac{\left(\text{Stroke Volume}_{\text{forward}}\right)}{\left(\text{Stroke Volume}_{\text{angiography}}\right)}$$

The angiographic stroke volume is estimated from the ventricular volumes obtained by angiography and is equal to the difference between end-diastolic volume and end-systolic volume. The forward stroke volume is calculated by dividing the cardiac output (determined by either the Fick or thermodilution method) by the heart rate. A regurgitant fraction of 0% to 20% represents mild aortic regurgitation, 20% to 40% represents moderate aortic regurgitation, and greater than 40% indicates severe aortic regurgitation. This method is rarely used clinically in the current era.

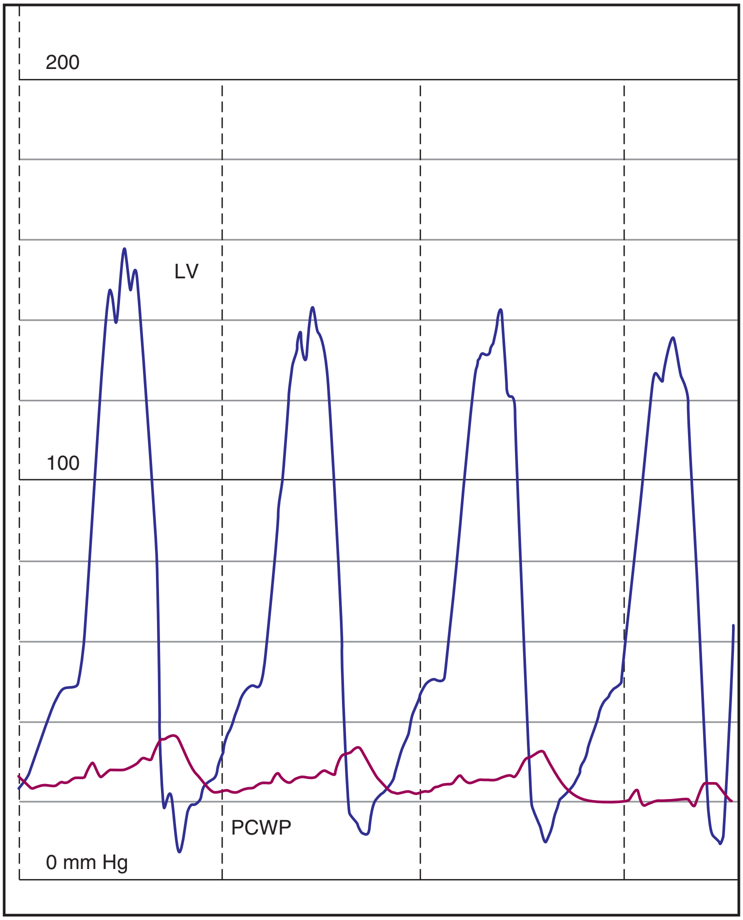


Fig. 6.7 Patient with severe acute aortic regurgitation and marked elevation in the left ventricular (LV) diastolic pressure exceeding the pulmonary capillary wedge pressure (PCWP).

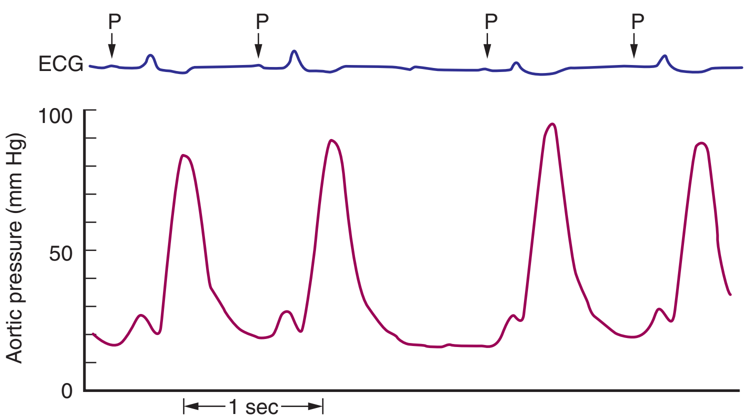


Fig. 6.8 Premature opening of the aortic valve in severe acute aortic regurgitation may result in transmission of a pressure wave from atrial contraction (P wave on electrocardiogram [ECG]) to the aorta and appears as an *a* wave on the aortic pressure trace, as shown here. (From Alexopoulos D, Sherman W. Unusual hemodynamic presentation of acute aortic regurgitation following percutaneous balloon valvuloplasty. *Am Heart J*. 1988;116:1622–1623.)

Box 6.2 Angiographic Grading of the Severity of Aortic Regurgitation

1. 1+ Small amount of contrast in the left ventricle during diastole that clears with each beat and never completely fills the left ventricular chamber
2. 2+ Faint opacification of the entire left ventricle
3. 3+ Dense opacification of the left ventricle (as dense as the aorta)
4. 4+ Complete and dense opacification of the left ventricle during the first cardiac cycle, and the left ventricle is more densely opacified than the aorta

AORTIC STENOSIS

Aortic valve stenosis is one of the most common valvular lesions in clinical practice. Adult patients who present with aortic stenosis at relatively younger ages (i.e., younger than 65 years) often have a congenitally bicuspid valve, whereas patients who present older than this age have calcific, tricuspid aortic valves (termed *calcific aortic stenosis* or *senile aortic stenosis*). Other disease processes rarely cause aortic valvular stenosis and include rheumatic heart disease, radiation-induced valvulitis, Paget disease of bone, and end-stage renal failure. It is important to distinguish aortic valvular stenosis from related conditions causing obstruction to left ventricular outflow, such as hypertrophic obstructive cardiomyopathy, subvalvular membranes, supravalvular aortic stenosis, or coarctation of the aorta.

Physiology of Aortic Valve Stenosis

The presence of a pressure gradient across the aortic valve defines aortic valvular stenosis. Normally, a small pressure gradient is present very early in systole when simultaneous left ventricular and aortic pressures are measured with sensitive, high-fidelity catheter-tipped micromanometers that represent an *impulse gradient* during the rapid phase of ventricular ejection.[5] The pathological gradient of aortic stenosis persists through systole. Progressive obstruction leads to pressure overload of the left ventricle and compensatory left ventricular hypertrophy with associated diastolic abnormalities from increased myocardial stiffness and reduced left ventricular compliance.

Aortic stenosis is progressive. The rate of progression is variable for any individual, but the annual increase in pressure gradient averages 7 mm Hg, and the annual decrease in aortic valve area (AVA) averages 0.1 cm^2.[6] Patients with aortic stenosis remain asymptomatic for prolonged periods until valve stenosis is severe. Symptoms include chest pain, syncope, and dyspnea. Interestingly, the severity of stenosis alone does not necessarily correlate with symptom status or the nature of symptoms. A study of patients with severe aortic stenosis and preserved ejection fraction found that the extent of obstruction (based on the mean pressure gradient and AVA) was the same regardless of the nature of symptoms (e.g., asymptomatic, syncope, chest pain, or dyspnea).[7] However, patients with syncope had smaller left ventricular cavities and reduced cardiac outputs, and patients with dyspnea had more profound diastolic abnormalities.[7] The cause of angina chest pain is primarily from increased wall stress and subendocardial ischemia from coronary blood flow supply-demand mismatch, although coronary artery disease is commonly present in this population as well. Sudden cardiac death occurs primarily in symptomatic patients with severe aortic stenosis; it is rarely observed in adult patients without symptoms.

Several complex factors impact the pressure gradient that exists between the left ventricle and the ascending aorta in patients with valvular aortic stenosis. The first set of factors relates to the complex nature of fluid mechanics and flow through a stenotic orifice (Fig. 6.9). Theoretical considerations of flow through a stenotic orifice predict the presence of intraventricular pressure gradients because of a drop in pressure from the body of the left ventricle to the outflow tract from a tapering of the flow field, with subsequent acceleration of blood flow as it approaches the stenotic orifice. This observation has been confirmed in patients with valvular aortic stenosis. Elegant investigations using micromanometer catheters to precisely measure the chamber pressure in patients with severe valvular aortic stenosis reveal that subvalvular gradients comprise nearly half of the total pressure gradient.[8] This observation is critically important to clinicians using cardiac catheterization to estimate the severity of aortic stenosis because failure to properly position the catheter deep within the left ventricle may incompletely estimate the true transvalvular gradient. Fluid dynamic theory also predicts the phenomenon of *pressure recovery*. The maximum pressure drop and the zone of minimal pressure and maximum velocity exist at the site of obstruction (vena contracta) with the development of turbulent flow and loss of energy. Distal to the obstruction, some of this energy is recovered with the reestablishment of laminar flow, and, consequently, an increase in pressure occurs from *recovery* of some of the pressure dropped across the stenotic valve. This phenomenon has been observed in patients with significant aortic stenosis using micromanometer catheters to carefully measure the pressure from the left ventricle to the ascending aorta.[8] The zone of minimal pressure occurred just above the aortic valve, and the zone of pressure recovery occurred higher in the ascending aorta. The magnitude of pressure recovery averaged 10 mm Hg. Although the fluid-filled catheters with multiple side holes used for clinical estimation of valve area may not detect pressure recovery, clinicians should be aware of this phenomenon and position the aortic catheter in the most proximal location.

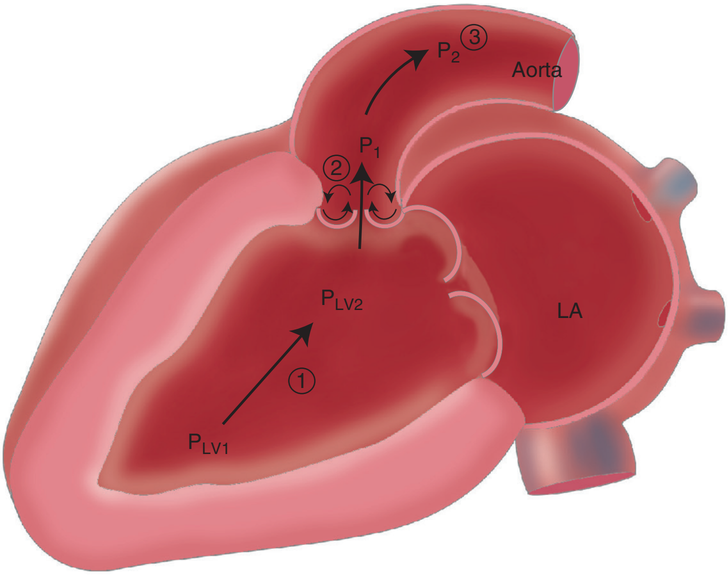


Fig. 6.9 Schematic diagram of the important factors involved in generation and interpretation of a pressure gradient. Intracavitary tapering of the flow field causes an intraventricular pressure gradient *(1)* where pressure at the apex *(P_{LV1})* exceeds the pressure below the aortic valve *(P_{LV2})*. Just after the site of obstruction, where maximum velocity (vena contracta) occurs *(2)*, is the site of minimal pressure. Turbulence in this area leads to pressure recovery *(3)*. Thus aortic pressure at P_1 is lower than aortic pressure at P_2. *LA*, Left atrium.

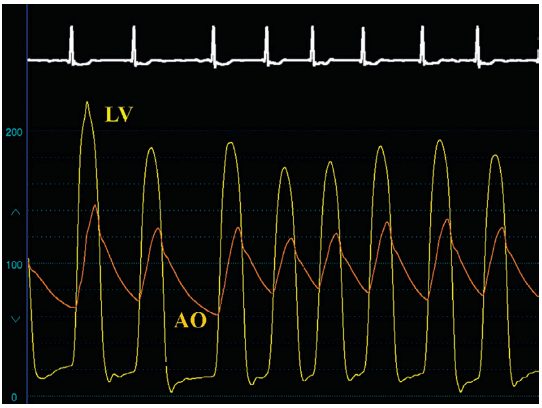


Fig. 6.10 Simultaneous left ventricular (LV) and aortic (AO) pressure tracings obtained in a patient with severe aortic stenosis and atrial fibrillation, demonstrating the varying systolic pressure gradients associated with varying RR intervals.

Neglecting the concept of pressure recovery and improper positioning of the aortic catheter when measuring a transvalvular gradient will lead to underestimation of the gradient and overestimation of the AVA with greater disparity observed with more severely narrowed aortic valves.[9]

The second set of factors that influences the pressure gradient relates to the fact that the pressure gradient is proportional not only to orifice area but also to flow across the valve. Flow will vary with the heart rate, contractile state of the heart, and the degree of preload and afterload. Thus flow may vary from beat to beat and with the respiratory cycle. Common arrhythmias, including marked sinus arrhythmia, atrial fibrillation with variations in the RR interval, and frequent premature atrial or ventricular beats or intermittent pacing, will affect the transvalvular pressure gradient (Fig. 6.10). For this reason the use of invasive techniques to estimate the severity of aortic valve stenosis mandates simultaneous left ventricular and aortic pressure measurements while maintaining the relative stability of these factors during data collection.

Determination of the Severity of Aortic Valve Stenosis

Estimating the severity of aortic stenosis often begins with noninvasive methods using echocardiographic techniques that utilize the Doppler methodology.[4] These well-established methods provide accurate estimations of the extent of valve narrowing by estimating the transvalvular gradients and aortic valve orifice area. Severe aortic stenosis is defined by a Doppler maximum aortic velocity of greater than or equal to 4.0 m/s, a mean transvalvular gradient greater than or equal to 40 mm Hg, and an estimated valve area less than or equal to 1.0 cm^2 or an AVA indexed to body surface less than or equal to 0.6 cm^2/m^2.[10] Understand that the definition of "severe" stenosis is somewhat arbitrary, and decisions regarding timing of valve replacement in a particular patient should not be constrained by fixation on a number.

The use of echocardiography to assess the severity of aortic stenosis has been well described elsewhere.[11,12] The peak, or maximum, instantaneous velocity across the aortic valve is measured by continuous wave Doppler (Fig. 6.11) and is an important direct measurement of the severity of aortic stenosis; a velocity of >4.0 m/s is consistent with severe aortic stenosis. The morphology of the continuous wave Doppler velocity envelope can provide helpful information regarding the severity of aortic stenosis. Mild and moderate degrees of aortic stenosis will have a rapid upstroke in the velocity envelope early in systole (see Fig. 6.11A), while patients with severe aortic stenosis will demonstrate a delay (see Fig. 6.11B). The peak velocity can be used to calculate a peak pressure gradient by using the simplified Bernouli equation as follows:

$$\text{Peak Pressure Gradient} = 4 \times (\text{velocity across the aortic valve})^2$$

Note that the peak pressure gradient describes the highest gradient that exists during the cardiac cycle. The mean gradient is calculated as the average of the instantaneous gradients occurring during systole; this value is generated by the echocardiographic imaging program and is lower than the peak pressure gradient; a mean gradient greater than or equal to 40 mm Hg is consistent with severe stenosis. Finally, the valve area is calculated using the continuity equation, which states that the volume of blood passing through the left ventricular outflow tract (LVOT stroke volume) is the same as the volume of blood passing through the aortic valve orifice (AV stroke volume). In turn, the stroke volume in a tube can be determined by the formula:

$$\text{Stroke volume} = (\text{Cross-sectional area of orifice}) \times (\text{Velocity time integral [VTI]})$$

Assuming a round orifice, the cross-sectional area of the LVOT can be calculated by:

$$\text{CSA LVOT} = \pi \times (\text{LVOT diameter}/2)^2$$

Because the continuity equation states that the AV stroke volume equals the LVOT stroke volume, the AVA can be calculated by the equation:

$$\text{AVA} = \frac{(\text{LVOT} \cdot \text{VTI}) \times (\text{LVOT} \cdot \text{CSA})}{\text{AV} \cdot \text{VTI}}$$

These echocardiographic measurements form the basis for our clinical decisions; in fact, invasive methods to determine the severity of stenosis are discouraged if noninvasive methods clearly show severe stenosis. In some cases, however, noninvasive techniques may be inconclusive or of inadequate quality or may provide data that are discordant with the clinical examination or a patient's symptoms. Pitfalls in the echocardiographic assessment of aortic stenosis severity include the inability to obtain an adequate continuous wave Doppler signal and improper measurement of the LVOT diameter. If the Doppler is not aligned with the ejected jet of blood, the peak velocity and therefore the aortic stenosis severity will be underestimated. Inaccurate measurement of the LVOT diameter is another common source of errors in valve area calculation. The stenosis severity will be overestimated if the LVOT diameter is erroneously measured too small, and the stenosis severity will be underestimated if the LVOT diameter is erroneously measured too large.[11–13]

When the degree of stenosis is unclear from echocardiography, clinicians turn to an invasive assessment.[10] There is evidence that in the current era the correlation between invasive and noninvasive determinations of stenosis severity is lower than previously thought because of variability in echo measurements, thus supplementary invasive data are often helpful in borderline cases.[14]

Invasive methodology applies an adaptation of the Gorlin formula to calculate the area of the stenotic aortic valve. This technique has been in clinical use for many years, is fairly simple to calculate from variables obtained during catheterization, and provides clinicians and patients an easy-to-visualize number (i.e., AVA in centimeters squared).

The Gorlin Formula for the Estimation of AVA

The mathematical derivation of the Gorlin formula to estimate the orifice area is based on the idealized physics of hydraulic systems and has been explained during the discussion of mitral valve stenosis (see Chapter 5). Gorlin solved the equation using the valve area of a stenotic mitral valve measured at autopsy from a single patient and determined

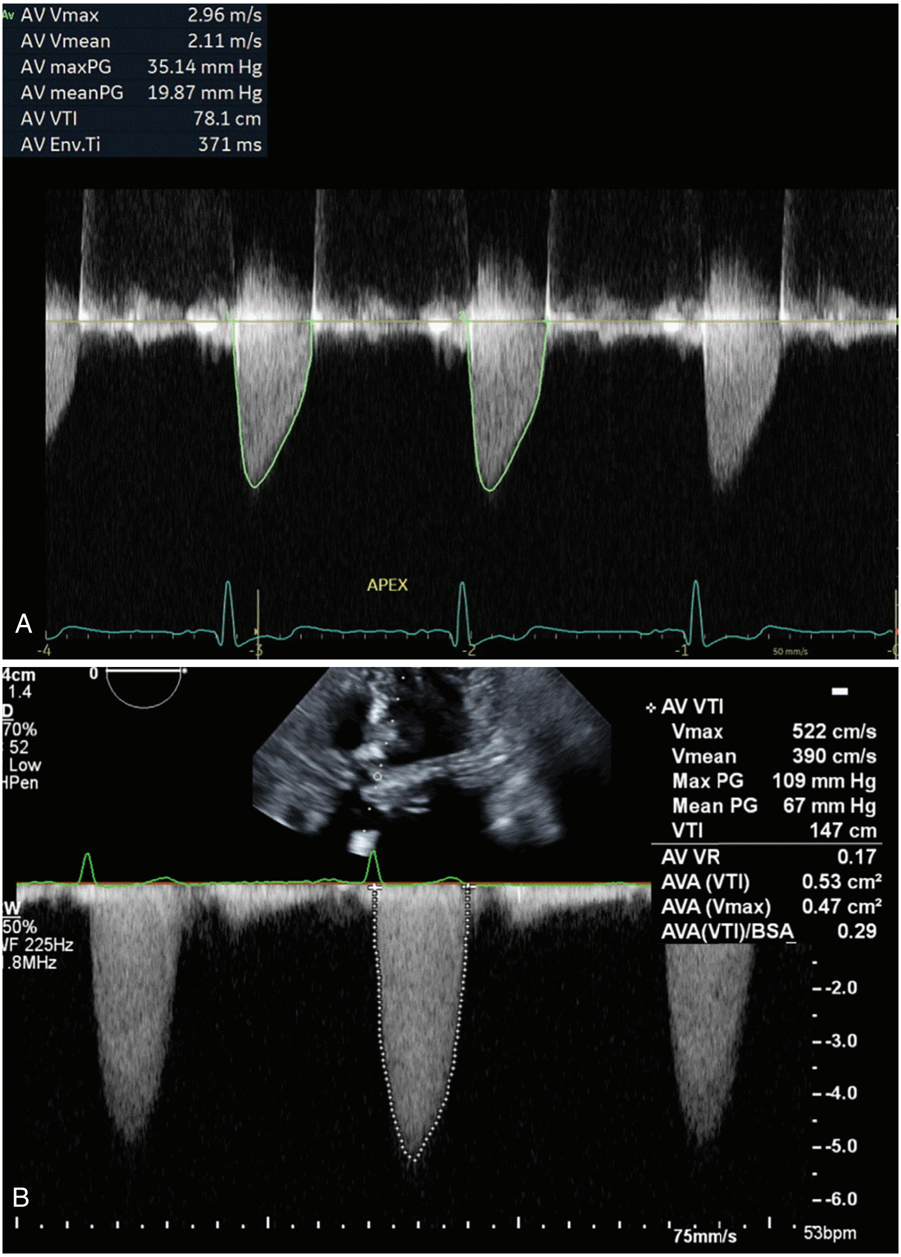


Fig. 6.11 Echocardiographic Doppler images from patients with aortic stenosis. (A) A patient with moderate aortic stenosis with a maximum velocity of 2.96 m/s. Note the rapid upslope of the Doppler profile consistent with moderate aortic stenosis. (B) A patient with very severe aortic stenosis and a maximum velocity of 5.22 m/s. Note the delayed upstroke in the Doppler profile. *AV*, Aortic valve; *AVA*, aortic valve area; *BSA*, body surface area; *PG*, peak gradient; *Vmax*, maximal velocity; *Vmean*, mean velocity; *VTI*, velocity time integral.

the value of the coefficient C to be 0.7. He then compared the calculated valve area with the area measured at either autopsy (six patients) or operation (five patients) for validation.[15] Gorlin noted that the formula could be adapted to the aortic valve by using the systolic ejection period (SEP) to account for the time that flow occurs across the aortic valve, but he did not know the value of the coefficient. This has subsequently never been determined. Nevertheless, the Gorlin formula applied to the aortic valve is generally used as follows:

$$\text{Aorticvalve area} = \frac{\text{Cardiac output}}{\text{Heart rate}\left(\text{SEP}\right)\left(44 \cdot 5\right)\text{C}\left(\sqrt{\text{P}_1 - \text{P}_2}\right)}$$

The coefficient C has traditionally been set at 1.0 (based on the assumption that the effective and anatomic orifice areas are identical for the aortic valve), with $P_1–P_2$ representing the mean transvalvular pressure gradient.

Although the Gorlin formula for the aortic valve is generally well established by the cardiology community and has been the basis of clinical decision-making in valvular heart disease for over a generation, several important criticisms potentially limit its use. Although the formula calculates the *valve area* in centimeters squared, the formula actually provides the *effective* orifice area rather than a determination of a true anatomic area. The formula is based on an idealized view of a stenotic valve and assumes that the stenotic orifice is round and the degree of narrowing is rigid and fixed. In fact, the usual stenotic aortic valve is a highly distorted orifice, and there often remains some degree of mobility that may vary the orifice area, depending on the flow across the valve. The coefficients used by the formula are not constant and may vary with both the size of the gradient and the cardiac output. In addition, the formula is not valid in the setting of significant regurgitation because the flow across the valve represents the summation of both the forward flow and regurgitant flow, the latter being difficult to quantify. Finally, the Gorlin formula depends on precise measurement of the transvalvular gradient, SEP, and cardiac output, each of which may include sources of error.

In the majority of patients with aortic stenosis, the portion of the Gorlin formula described in the denominator as *heart rate* × *SEP* × *44.5* calculates to around 1000. Thus the Gorlin formula can be simplified by the formula described by Hakki et al.[16]:

$$\text{Aortic valve area} = \frac{\text{Cardiac output in liters per minute}}{\sqrt{\text{Pressure gradient}}}$$

In a study of 60 patients with varying degrees of aortic stenosis, the calculated AVA using the Hakki[16] formula correlated well with the valve area determined by the more complex Gorlin formula even when the peak gradient was substituted for mean gradient. The observed difference between the two calculations rarely exceeded 0.2 cm^2, especially in patients with valve areas less than 0.7 cm^2, providing confidence that a designation of *severe* aortic stenosis would not change depending on the formula used. Although valve areas in most patients with moderate degrees of aortic stenosis were within 0.2 cm^2, the observed difference in valve areas between the two formulas was as high as 0.42 cm^2, suggesting that the more detailed Gorlin formula may be more appropriate to correctly classify the stenosis severity in patients with borderline degrees of stenosis.

Application of the Gorlin Formula for the Estimation of AVA

The Gorlin formula for calculation of the AVA requires the measurement of cardiac output, SEP, heart rate, and mean transvalvular pressure gradient. Cardiac output is easily and routinely measured in the cardiac catheterization laboratory, using either thermodilution or Fick methods; however, it should be emphasized that great care must be taken in making this determination because small differences in cardiac output translate to potentially important changes in the calculated valve area. Similarly, variations in measurement of the SEP may result in differences in valve areas. The SEP begins when the intraventricular pressure rises above the aortic pressure. The end of the SEP has been variably defined as either the second left ventricular aortic pressure crossover point or at the aortic incisura (Fig. 6.12). Carefully performed hemodynamic studies have shown that the aortic incisura more accurately reflects the end of aortic ejection rather than the second left ventricular aortic pressure crossover point.[17]

Modern-day catheterization laboratories outfitted with computerized physiologic recording systems provide automated calculations of the AVA, obviating the need for physicians to recall the formula, to manually measure specific variables or perform the calculations. The required elements of the Gorlin formula are either directly entered into the computer (e.g., cardiac output) or obtained from computerized analysis of the waveforms (Fig. 6.13). These automated systems are usually reliable; however, the careful physician should always review the computerized analysis for errors in measurement of the gradient, heart rate, or SEP, particularly if the waveforms or surface electrocardiogram contains artifacts or is of poor quality. Variations in the heart rate or changing pressure gradients, as observed in atrial fibrillation, or marked respiratory variations, require an analysis of at least 10 consecutive beats for the most accurate estimation of valve area.

Measurement of the Transvalvular Pressure Gradient

Confusion often occurs regarding the various terms used to describe the pressure gradient (Fig. 6.14). The *peak-to-peak* gradient is the difference between the peak left ventricular and aortic systolic pressure and is often used to quickly report the transvalvular gradient in the cardiac catheterization laboratory, but it has no physiologic relevance because these peaks occur at different times. The *peak instantaneous* gradient is obtained by Doppler echocardiography and represents the largest gradient that exists between the left ventricle and the aorta. The peak instantaneous gradient is greater than the peak-to-peak gradient. Finally, the *mean* gradient, described by both echocardiography and catheterization, should be similar in magnitude. The mean gradient is used in the Gorlin formula and represents

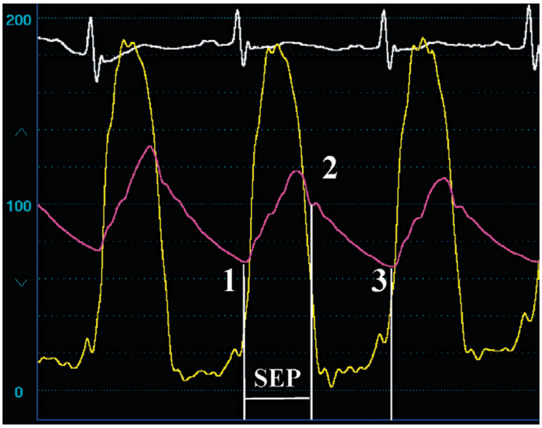


Fig. 6.12 Determination of the systolic ejection period (SEP) for calculation of the aortic valve area using the Gorlin formula. Systole begins when the left ventricular and aortic pressures first cross *(point 1)* and end at the dicrotic notch *(point 2)*. The *SEP* represents the time from *point 1* to *point 2* in seconds. The cardiac cycle ends at *point 3*.

Fig. 6.13 Example of an automated computerized analysis for the determination of several of the variables required to calculate the aortic valve area with the Gorlin formula. In this case the computer determined a mean systolic gradient *(white-shaded area)* of 32.4 mm Hg, a heart rate of 86 beats per minute, and a systolic ejection period of 0.394 s. The cardiac output measured is 4.3 L/min or 4300 mL/min. Thus the calculated aortic valve area is 0.5 cm^2.

the average of all the instantaneous gradients that exist during systole. Clinicians typically report the mean gradient to describe the severity of aortic stenosis.

Clinicians have employed numerous methods to measure the transvalvular pressure gradient in the cardiac catheterization laboratory (Box 6.3).[18] Many are fraught with the potential for significant error. The ideal method takes into account the presence of intraventricular pressure gradients and the phenomenon of pressure recovery in the ascending aorta (described earlier) and avoids positioning a catheter across the aortic valve to record the left ventricular pressure,

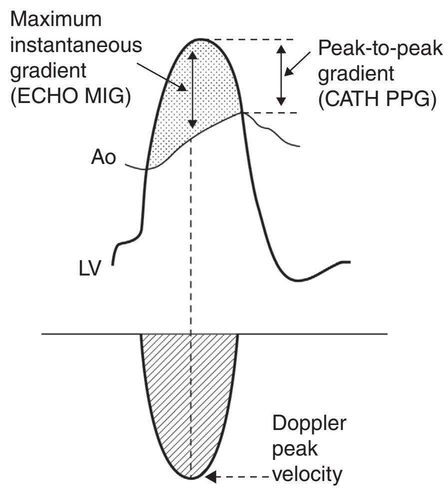


Fig. 6.14 Demonstration of the relationship between the maximum instantaneous gradient obtained by echocardiography, the peak-to-peak gradient obtained by cardiac catheterization, and the Doppler peak velocity obtained by echocardiography. *Ao*, Aorta; *CATH PPG*, cardiac catheterization peak-to-peak gradient; *ECHO MIG*, echocardiographic maximum instantaneous gradient; *LV*, left ventricle. (From Abbas AE, Pibarot P. Hemodynamic characterization of aortic disease states. *Cathet Cardiovasc Interv.* 2019;93:1002–1023.)

Box 6.3 Methods of Measuring the Transvalvular Gradient in Aortic Stenosis

LV via transseptal; AO catheter retrograde above AO valve
LV retrograde with pressure wire; AO catheter retrograde above AO valve
LV retrograde with pigtail; AO catheter retrograde above AO valve
LV and AO retrograde with dual lumen pigtail
LV retrograde with pigtail; AO pressure from the sidearm of long sheath
LV retrograde with pigtail; AO pressure from the sidearm of femoral sheath
LV retrograde with pigtail and “pullback” pressure from LV to AO

thereby preventing additional orifice obstruction by the profile of the catheter itself.[19] This ideal catheter configuration can be achieved by simultaneously recording pressure from two catheters: one positioned directly above the aortic valve from a retrograde approach via the femoral, radial, or brachial arteries and one placed by a transseptal approach from the femoral vein, across the atrial septum and mitral valve, and into the body of the left ventricle toward the apex (Fig. 6.15). This technique may also alleviate the concern regarding the risk of cerebral embolic events from retrograde crossing of the stenotic aortic valve.[20,21] However, transseptal catheterization entails additional risk, and many physicians who perform routine diagnostic catheterization lack transseptal catheterization skills. Thus this ideal catheter arrangement is not commonly used unless the aortic valve cannot be crossed by a retrograde approach.

Another method employs a 0.014-inch pressure wire (designed for the measurement of intracoronary pressure) positioned in a retrograde manner across the aortic valve to measure the left ventricular pressure while simultaneously measuring the aortic pressure from a catheter placed just above the aortic valve.[18] Advantages of this method include the extremely low profile of the pressure wire across the valve, preventing the potential for additional obstruction from a catheter, the generation of high-fidelity waveforms from the micromanometer near the wire tip, and the requirement of only a single arterial access site to obtain these data (Fig. 6.16). A major disadvantage relates to the significant additional cost of the pressure wire compared with the relatively inexpensive table-mounted transducers. In addition, the waveform data from the pressure wire may not easily feed into the catheterization laboratory computerized systems resulting in difficulty in determining the variables needed to calculate the valve area.

Simultaneous left ventricular and aortic pressures can be accurately measured from a retrograde catheter placed across the aortic valve into the left ventricle and a second catheter positioned from another arterial site in the ascending aorta just above the valve. This arrangement provides high-quality hemodynamic data but is not very popular because it requires two arterial punctures, thereby potentially increasing the risk of a vascular complication. In addition, the presence of a catheter across the aortic valve may contribute additional obstruction, although this effect is usually small.

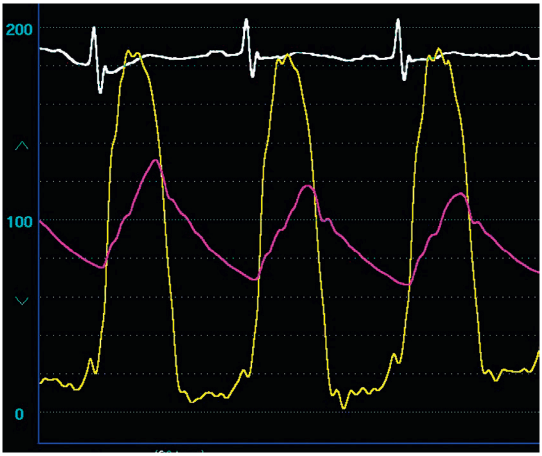


Fig. 6.15 Simultaneous left ventricular and central aortic pressures in a patient with severe aortic stenosis. To avoid the presence of a catheter across the aortic valve, which may contribute to additional obstruction to flow, the left ventricular pressure was recorded from a catheter placed into the left ventricle by a transseptal approach. Central aortic pressure was recorded from a catheter positioned just above the aortic valve from a retrograde approach via the femoral artery. Note the marked delay in upstroke, presence of a crisp dicrotic notch, and lack of time delay on the central aortic pressure. This method represents the ideal technique for recording the transvalvular pressure gradient.

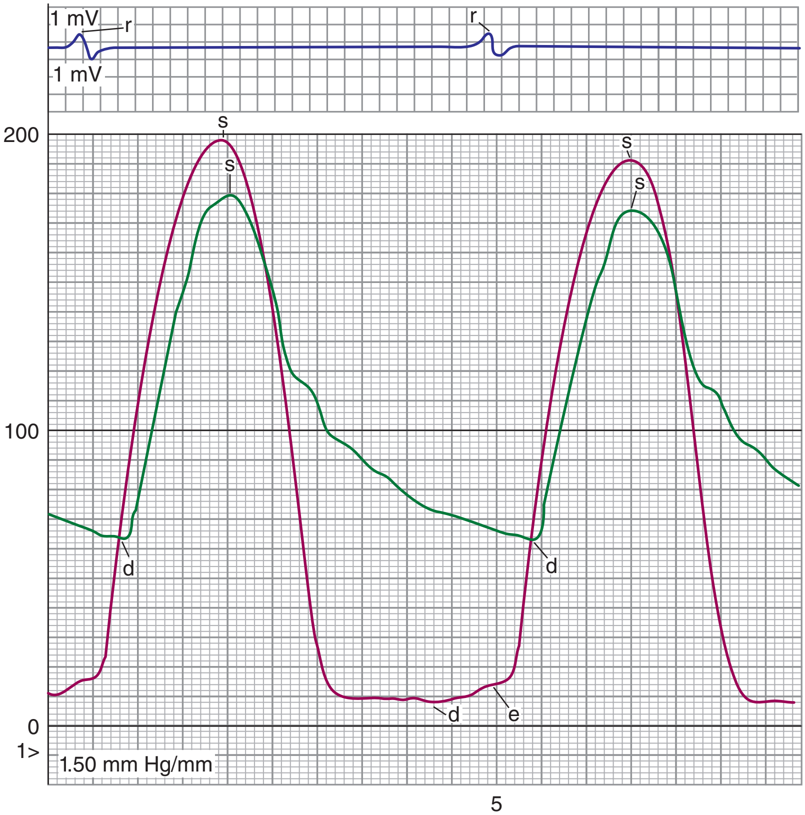


Fig. 6.16 Simultaneous left ventricular and central aortic pressures in a patient with mild aortic stenosis. The left ventricular pressure was obtained with a 0.014-inch pressure wire, and the central aortic pressure was obtained from a catheter positioned above the aortic valve using a retrograde approach via the femoral artery. *d*, Diastolic pressure; *e*, end-diastolic pressure; *r*, R wave on electrocardiogram; *s*, systolic pressure.

The dual lumen pigtail catheter (or Langston catheter) allows simultaneous pressure measurements above and below the valve from a single arterial access site, thereby avoiding the need for two arterial punctures. Criticism of earlier versions of this catheter centered on the relatively small size of the aortic pressure lumen, resulting in dampened waveforms and loss of fidelity. Current, 6-French versions (Vascular Solutions, Minneapolis, Minnesota) of this catheter perform well (Fig. 6.17), and although slight damping of the aortic pressure wave can be seen, it has little clinical significance. Available in several French sizes (6–8) and shapes (pigtail, multipurpose A2, and straight), the aortic side holes lie about 8 cm from the end of the catheter, allowing positioning well within the ventricle and just above the aortic valve. If a dual lumen catheter is not available, simultaneous left ventricular and ascending aortic pressures can be collected from a single arterial puncture by using a "mother and daughter" catheter configuration. A short (90 cm), 6- to 8-French guide catheter is used to measure the aortic pressure by attaching a hemostatic valve outfitted with a sidearm for pressure measurement. The left ventricular pressure is measured by inserting a 4-French, longer (110 cm or longer) catheter inside the guide catheter.

Some physicians favor the use of a retrograde catheter placed across the aortic valve to measure the left ventricular pressure while simultaneously recording the femoral artery sheath sidearm pressure as a surrogate for the central aortic pressure. This technique requires using a sheath at least 1-French size greater than the pigtail catheter. This arrangement is fraught with potential error and should not be used to measure the pressure gradient. The error is most likely to have a clinical impact when there is stenosis of moderate severity and when there is marked discrepancy between the femoral artery sheath and central aortic pressure. An important source of error relates to the temporal lag between the central aortic and peripheral artery pressure; this time delay in the femoral artery sheath relative to the left ventricular pressure falsely raises the mean transvalvular pressure gradient. An additional source of error is due to peripheral amplification, resulting in a higher systolic pressure in the femoral artery compared with the central aorta. This will falsely lower the mean transvalvular pressure gradient. Fig. 6.18 provides an example of the typical effect of temporal delay and systolic pressure amplification observed with this method. Although these two errors have opposing effects and may actually negate each other, potentially impacting little on the valve area calculation, this method is discouraged and should be avoided. The temptation to correct for the time delay by phase shifting the femoral artery sheath tracing to the left should be resisted because this will leave the effect of peripheral amplification unopposed and will actually exacerbate the error. Another source of error with this approach occurs when the femoral artery sheath pressure records lower than the central aortic pressure; this occurs when there is an iliac stenosis from peripheral vascular disease, vessel tortuosity, or when the sheath clots or kinks. These common situations falsely raise the gradient, overestimating the severity of aortic stenosis. A long (55 cm) sheath positioned distal to the origin of the subclavian artery minimizes these effects.[22] Again, because of these multiple sources of error, the left ventricular-arterial sheath sidearm method should be avoided.

The final technique for measuring transvalvular gradient relies on a recording of pressure during *pullback* from the left ventricle to the aorta. This method is simply not a valid means for measurement of the pressure gradient in patients with aortic stenosis and should be used only as a method of screening for the presence of a gradient in patients who undergo cardiac catheterization. The effects of the respiratory cycle, ventricular ectopy, and rhythm irregularity greatly impact the pressure gradient; accurate estimation of the gradient mandates simultaneous pressure recordings (Fig. 6.19).

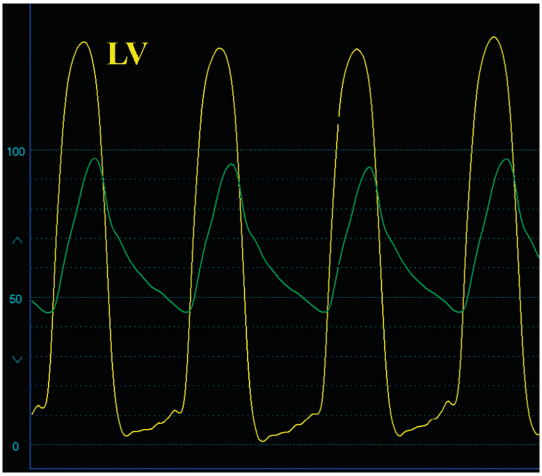


Fig. 6.17 Simultaneous left ventricular (LV) and central aortic pressures collected from a dual lumen (Langston) catheter in a patient with severe aortic stenosis.

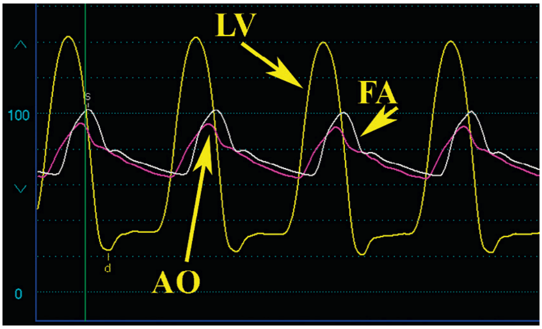


Fig. 6.18 Simultaneous recordings from the sidearm of a 7-French femoral arterial (FA) sheath and a dual lumen pigtail. The central aortic and left ventricular (LV) pressures were recorded in a patient with severe aortic (AO) stenosis. Note the significant time delay and systolic pressure amplification apparent on the femoral artery sheath pressure. An overestimation of the gradient occurs because of the time delay, as well as an underestimation because of peripheral amplification. The effects of these two opposing errors may, in fact, come close to canceling each other out and would not likely change the valve area calculation in most cases of severe aortic stenosis. However, this error would have more potential impact in borderline cases. Although phase shifting can correct the time delay, it will not correct the effect of peripheral amplification and would only exacerbate the error, leaving the effect of peripheral amplification unopposed.

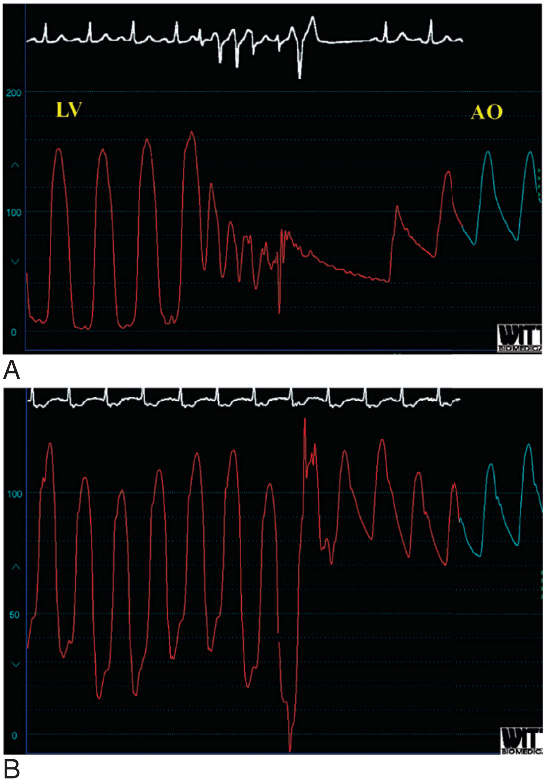


Fig. 6.19 Recording of pressure during a pullback from the left ventricle to the aorta. (A) The first recording is from a patient with mild aortic (AO) stenosis and shows substantial ventricular ectopy during this maneuver. (B) The second recording shows marked respiratory variation. Determination of the presence or extent of a pressure gradient without simultaneous pressure recordings in these cases is not possible. *LV*, Left ventricular.

The various techniques used to measure the transvalvular gradient have been evaluated.[23,24] Compared with simultaneous left ventricular and central aortic pressure, the left ventricle to the aorta pullback method underestimates the degree of stenosis.[23] One study compared the pressure gradient in 15 patients with aortic stenosis from eight different catheter configurations.[24] All eight catheter configurations measured the left ventricular pressure by a retrograde approach from two positions within the left ventricle: the body of the left ventricle near the apex and the outflow tract just below the aortic valve. For aortic pressure, the study compared four different methods: the ascending aorta at the level of the coronary arteries, ascending aorta 5 cm distal with the valve, femoral artery sheath pressure unadjusted for the time delay, and femoral artery sheath pressure adjusted (i.e., realigned) for the time delay. Compared with the gradients measured from catheters in the body of the left ventricle and the aorta at the level of the coronary arteries, all other positions underestimated the gradient. Most of the difference was attributable to the presence of an intraventricular gradient rather than pressure recovery, stressing the importance of positioning the catheter in the body of the left ventricle instead of in the LVOT. Furthermore, the use of an aligned femoral artery sheath pressure was associated with substantial underestimation of the valve area. Importantly, patients with moderate aortic stenosis exhibited the maximum variation (0.3 cm^2) and are a group in whom this degree of difference in valve area may change the management decision.

To summarize, the transvalvular gradient is an important variable for determination of the severity of aortic stenosis, and various methods to obtain this measurement are possible, with some having more pitfalls than others. After considering all these methods and balancing cost, patient safety, and accuracy, it is our practice at the University of Virginia to measure this gradient from a single arterial puncture using a dual lumen catheter.

Hemodynamic Findings in Aortic Stenosis

The calculated valve area is often used to assign the severity of aortic stenosis. The designations are somewhat arbitrary; nevertheless, a valve area greater than 1.5 cm^2 and a mean gradient less than 20 mm Hg are classified as mild aortic stenosis, and a valve area between 1.0 and 1.5 cm^2 with a mean gradient between 20 and 39 mm Hg is characterized as moderate aortic stenosis. In these patients, symptoms are rare without other heart disease or symptoms are seen only during stress, such as fever, extreme exertion, or tachyarrhythmia. A mean gradient greater than or equal to 40 mm Hg and a calculated valve area less than or equal to 1.0 cm^2 represent *severe* aortic stenosis, with symptoms associated with rest or minimal exertion. The term *critical* aortic stenosis is somewhat of an emotional term that terrifies patients; nevertheless, the term *very severe* aortic stenosis is used to describe patients with peak velocities on echo greater than or equal to 5.0 m/s and/or mean gradients in excess of 60 mm Hg. Clinicians must remain mindful of the limitations in valve area calculation and not base decisions solely on a number but take into account the patient's symptoms, ventricular function, compensatory processes, physical examination, and results of noninvasive tests, in addition to the calculated valve area.[4]

Mild aortic stenosis results in small transvalvular gradients, normal aortic upstroke, and normal right-sided pressures. With severe aortic stenosis, a variety of interesting hemodynamic findings are possible. The peak-to-peak transvalvular systolic pressure gradient generally exceeds 50 mm Hg and may top 100 mm Hg. In cases of severe aortic stenosis, when a retrograde approach is used to obtain the left ventricular pressure, the profile of the catheter may contribute to obstruction, adding to the transvalvular gradient. This may be observed as a rise in the aortic pressure of at least 5 mm Hg when the catheter is withdrawn from the left ventricle and is known as the *Carabello sign*, after the original observer of this phenomenon who noted this occurrence in patients with a valve area less than 0.6 cm^2 (Fig. 6.20).[25]

One of the cardinal features of severe aortic stenosis is the marked delay in upstroke on the aortic pressure waveform. Additionally, a prominent anacrotic notch or shoulder may appear within the upstroke from turbulent flow across the valve (Fig. 6.21). Pulsus alternans is a rare finding in patients with severe aortic stenosis and is usually associated with heart failure and poor left ventricular function (Fig. 6.22).[26]

Although right-sided pressures are usually normal unless there is heart failure, pulmonary hypertension may be seen in up to a third of patients with severe aortic stenosis and portends a poor prognosis, although it generally improves after aortic valve replacement.[27,28] It is not uncommon to also observe mitral valve gradients in association with severe aortic stenosis. These are usually due to the severe, mitral annular calcification seen in association with

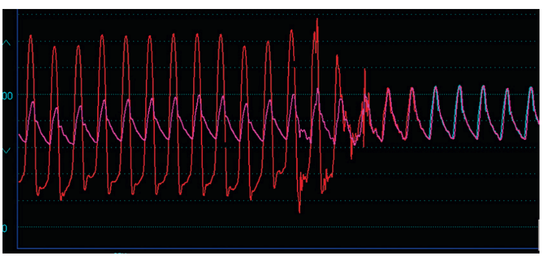


Fig. 6.20 Example of the Carabello sign in severe aortic stenosis. During pullback of a pigtail catheter from the left ventricle to the aorta, the systolic pressure in the aortic waveform increases with the relief of additional obstruction caused by the profile of the catheter.

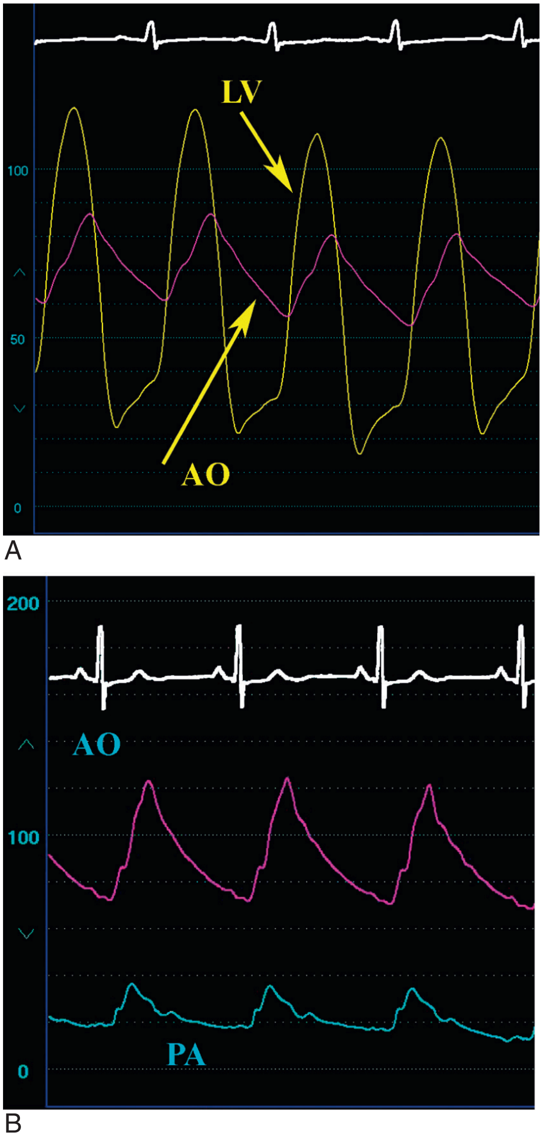


Fig. 6.21 Example of the marked delay in upstroke on the aortic (AO) pressure tracing in a patient with severe aortic stenosis. (A) Simultaneous left ventricular (LV) and AO pressure. (B) AO and pulmonary artery (PA) pressure. Note the anacrotic notch on the upstroke of the aortic pressure tracing.

calcific aortic stenosis. The typical gradient is in the mild-to-moderate category, averaging about 5 mm Hg (Fig. 6.23), but this finding is associated with diastolic abnormalities and higher left atrial and pulmonary arterial pressure.[29]

Discordance Between the Pressure Gradient and Valve Area

When the various parameters defining the severity of aortic stenosis are concordant, clinical decisions are straightforward. However, it is not uncommon that we observe discordance among these parameters. The most common cause of discordance relates to errors in echocardiographic measurement of the valvular or LVOT velocities usually because of an improper continuous wave Doppler signal, or because of mismeasurement of the LVOT diameter.[11,12] However, even

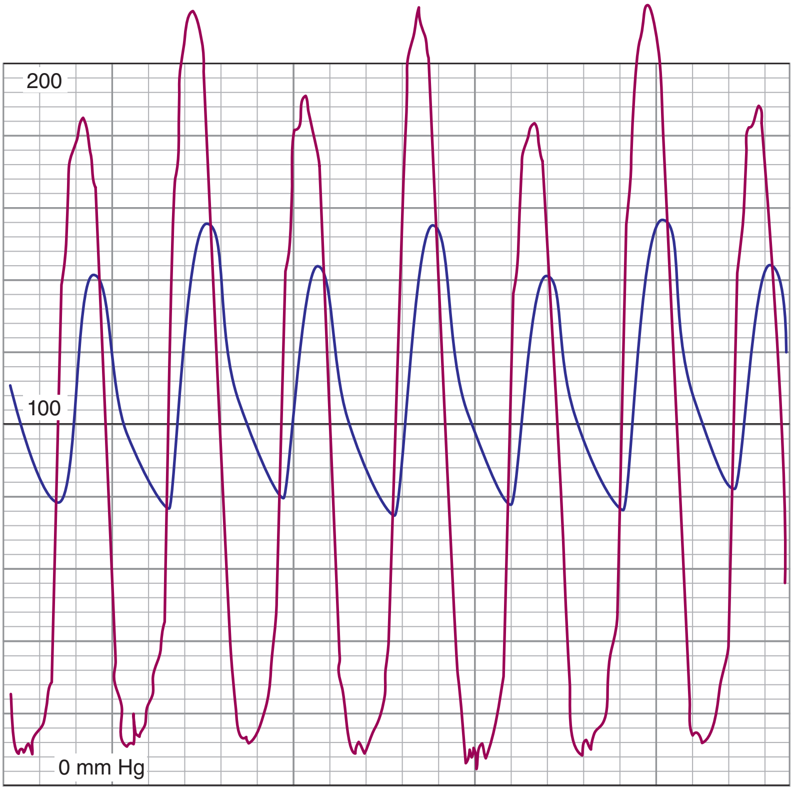


Fig. 6.22 Pulsus alternans in a patient with severe aortic stenosis.

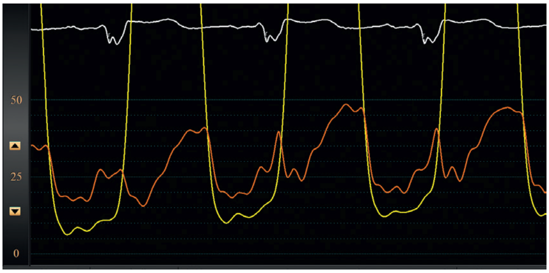


Fig. 6.23 Patients with calcific aortic stenosis often have severe mitral annular calcification that can cause gradients across the mitral valve. These usually are mild to moderate in severity, with mean gradients of less than 5 mm Hg as shown here, with simultaneous left ventricular and left atrial pressure.

if there are no measurement errors, there are several scenarios where there is discordance between the parameters resulting in physician head scratching and much consternation.

Once scenario occurs when the peak velocity is greater than 4.0 m/s and the mean gradient exceeds 40 mm Hg yet the calculated valve area is greater than 1.0 cm^2. This may be observed in the setting of increased flow across the aortic valve as seen in aortic insufficiency, large volume arteriovenous shunts (e.g., hemodialysis fistula), fever, tachycardia, hyperthyroidism, and anemia.

More commonly, the peak velocity is less than 4.0 m/s, the mean gradient is less than 40 mm Hg, but the calculated valve area is less than 1.0 cm^2. This scenario is termed "low-gradient aortic stenosis," and it is critical that the clinician determine if such patients simply have moderate aortic stenosis or if they truly have severe aortic stenosis. Because the pressure gradient is dependent not only on the valve area but also on the flow across the valve, pathological conditions resulting in low flow will result in low gradients despite severe aortic stenosis. Such patients will benefit from valve replacement. Alternatively, if there is normal flow, the low gradient indicates moderate aortic stenosis suggesting no benefit from valve replacement.

The phenomenon of "low-flow, low-gradient aortic stenosis" was initially described in patients with severe ventricular dysfunction. A low gradient (<30 mm Hg) in the setting of severe left ventricular dysfunction may indicate one of two conditions, each managed quite differently. In one scenario, the patient truly has severe aortic stenosis but is only able to generate a low-pressure gradient because of diminished cardiac output caused by pump failure. This patient will benefit from valve replacement. Alternatively, the patient may have severe left ventricular dysfunction and reduced cardiac output because of an underlying cardiomyopathy. The reduced contractile force causes a low cardiac output. The subsequent low-flow state in this patient may not allow even a mildly stenotic aortic valve to open adequately, giving a false impression of severe aortic stenosis (*pseudo*-severe aortic stenosis). The valve area is not fixed but is dependent on flow; this patient will not benefit from valve replacement as the underlying problem is primarily due to myocardial dysfunction.

In patients with reduced ejection fraction and low-gradient aortic stenosis, augmentation of flow with inotropic agents such as dobutamine can distinguish these two scenarios and help with decision-making. Dobutamine is infused until there is a >20% increase in cardiac output or stroke volume. If dobutamine infusion increases the pressure gradient to more than 40 mm Hg with no change in the valve area, true severe aortic stenosis is confirmed. Instead, if dobutamine increases the flow but does not change the gradient, the valve area will correspondingly increase and the diagnosis is pseudo-severe aortic stenosis. Valve replacement in this patient with heart failure caused primarily by a cardiomyopathy and not severe aortic stenosis will not lead to improved symptoms or ventricular function, and prognosis is generally poor with valve surgery.[30] The methodology of dobutamine infusion in distinguishing these subsets and indicating appropriate therapy has been described.[12,13,31–33]

The other subset of patients with low-gradient aortic stenosis and severe stenosis by valve area have preserved left ventricular function. In these cases, the clinician needs to distinguish patients with normal flow and low gradients from those with low flow and low gradients; the first group have moderate aortic stenosis, while the latter have severe aortic stenosis. Patients with "low-flow, low-gradient, severe aortic stenosis" have low gradients despite severe aortic stenosis because of a low stroke volume usually caused by a small ventricular cavity with restrictive physiology from severe hypertrophy. They are also known as "paradoxical low-flow, low-gradient aortic stenosis."[13,34,35] These patients may be misdiagnosed with moderate aortic stenosis, and their prognosis is poor if untreated. The typical patient with this condition is an elderly female with small body habitus, long-standing hypertension with severe hypertrophy, and a small ventricular cavity with a heavily calcified valve but a low gradient. The echocardiographic measurement of stroke volume is necessary to identify this subset with a stroke volume index less than 35 mm/m^2 considered low flow.[12,13] The role of dobutamine infusion in these patients is unclear since their ejection fraction is preserved or even hyperdynamic, and it is felt that dobutamine is unlikely to augment flow, although some experts do advocate for this. Perhaps more helpful is determination of the aortic valve calcium score by computerized tomographic (CT) imaging. Patients with low-flow, low-gradient aortic stenosis are unlikely to have severe stenosis if the calcium score is less than 1600 in males and less than 800 in females.[12]

A challenging clinical scenario may be observed in a patient with low-gradient aortic stenosis with normal left ventricular function and systemic hypertension. Such patients may be markedly dyspneic due to diastolic abnormalities from the left ventricular function and have elevated LVEDPs and subsequent pulmonary hypertension.[36] In these patients with marked systemic hypertension, there is increased afterload causing lower stroke volume and potential inaccuracies in determining the severity of aortic stenosis; usually the severity of stenosis is overestimated.[37] Determining if these individuals benefit from aortic valve replacement may prove difficult. Nitroprusside infusion during an invasive assessment of aortic stenosis has been shown to be a useful approach in such patients (Fig. 6.24). One study using nitroprusside infusion during invasive evaluation in the cardiac catheterization laboratory reclassified 25% of patients initially thought to have severe stenosis to moderate stenosis.[38]

Table 6.2 summarizes the hemodynamics of the various subsets of patients with low-gradient aortic stenosis.

Coexisting Aortic Regurgitation

Significant aortic regurgitation complicates the invasive assessment of aortic stenosis severity, primarily because the Gorlin formula requires knowledge of transvalvular flow during systole. The Fick and thermodilution methods assess only the effective forward flow, whereas transvalvular flow in mixed aortic regurgitation and stenosis includes regurgitant and forward flow. Using the Fick or thermodilution methods will *underestimate* total flow, providing an *overestimation* of the severity of the stenosis. There is no consensus on how to correct for this or how to report the results. Many clinicians report that the true valve area is *no less than* that calculated by the Gorlin formula. This may reassure the clinician in cases of moderate or mild aortic stenosis but is not helpful if the calculation suggests severe aortic stenosis.

Prosthetic Aortic Valves

The normally functioning prosthetic valve often results in some degree of obstruction. The degree of obstruction depends on the type and size of prosthesis. Expected gradients for surgical aortic valves are available and average 0

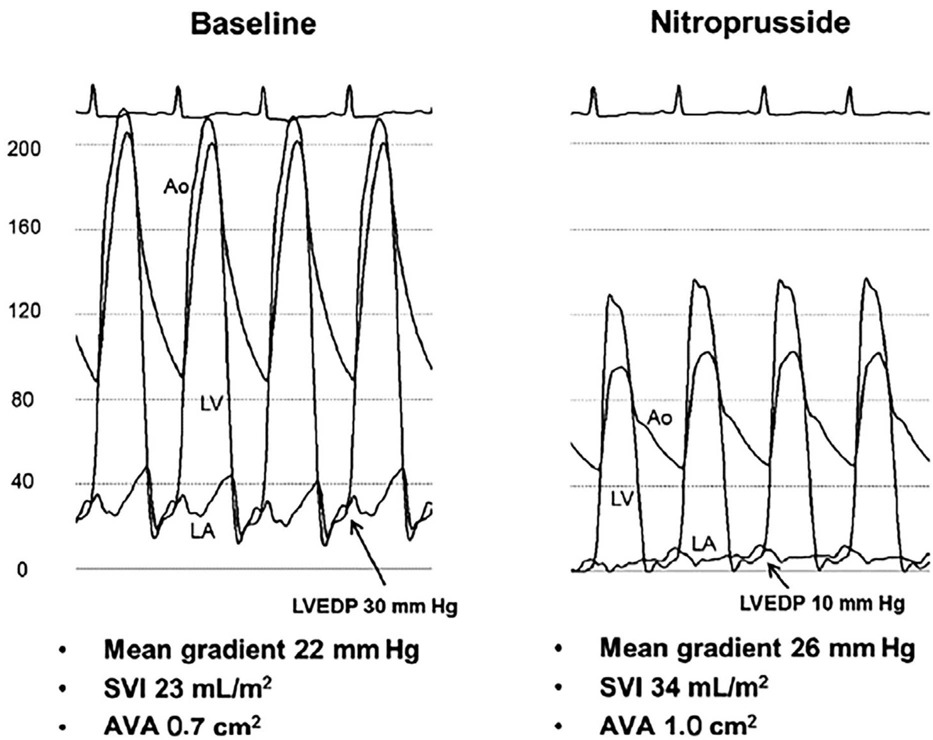


Fig. 6.24 Example of a patient with low-gradient aortic stenosis and preserved ventricular function in the setting of severe hypertension. The panel on the left shows severe hypertension with elevated left atrial and left ventricular end-diastolic pressure (LVEDP) with an associated low stroke volume and calculated valve area of 0.7 cm^2. After nitroprusside, although the gradient increases, the stroke volume also increased and the calculated valve area is larger at 1.0 cm^2. *Ao* , Aortic pressure; *AVA*, aortic valve area; *SVI*, stroke volume index. (Reproduced with permission from Eleid MF, Nishimura RA, Sorajja P, Borlaug BA. Systemic hypertension in low-gradient severe aortic stenosis with preserved ejection fraction. *Circulation.* 2013;128:1349–1353.)

Table 6.2 Categories of Low-Gradient Aortic Stenosis

	MODERATE AS	PSEUDO- AS	TRUE SEVERE AS	PARADOXICAL LOW-FLOW, LOW-GRADIENT AS
LV function	Normal	Reduced	Reduced	Normal
Stroke volume	Normal	Reduced	Reduced	Reduced
Cavity size	Normal	Enlarged	Enlarged	Small
Gradient response to dobutamine	Not needed	No change	Increase	Increase

AS, Aortic stenosis; *LV*, left ventricular.

to 15 mm Hg for bioprosthetic and modern mechanical valves in the aortic position. Gradients with archaic and much higher profiles such as the Starr-Edward valve could be as high as 40 mm Hg depending on the size.

If an invasive assessment of aortic valve prosthesis is performed, there is no value to calculation of valve area; most clinicians simply report a pressure gradient. One study of 135 patients with normally functioning prosthetic aortic valves compared the Gorlin-derived valve area with the actual orifice area of the prosthesis and, in many cases, found poor correlation with both overestimation and underestimation by greater than 0.25 cm^2.[39]

Similar to the native aortic valve, the assessment of the aortic bioprosthesis is best accomplished by echocardiography. This topic is complex and beyond the scope of this chapter and has been extensively reviewed elsewhere.[40] Nevertheless, invasive hemodynamics are sometimes required. Measurement of left ventricular pressure in patients with prosthetic valves requires special attention. Bioprosthetic valves can be crossed retrograde using similar methods as for native valves; mechanical valves cannot be crossed with conventional catheters in a retrograde manner because of potential catheter entrapment or propping open of a leaflet, causing acute severe regurgitation, which may not be tolerated. Traditionally, transseptal puncture has been used to measure the left ventricular pressure in this setting. The use of a 0.014-inch pressure wire to measure the left ventricular pressure by a retrograde approach in patients with mechanical aortic valve prosthesis is safe and reliable.[41] However, an *ex vivo* model has shown that some degree of aortic regurgitation occurs with a St. Jude valve (St. Jude Medical, St. Paul, Minnesota), Medtronic-Hall valve (Medtronic, Minneapolis, Minnesota), or Bjork Shiley valve.[42] An example is shown in Fig. 6.25.

Higher than expected gradients across prosthetic valves are concerning and may be clues to prosthetic valve dysfunction. Bioprosthetic valves degenerate over time and can develop severe stenosis. Mechanical valves are not prone to degeneration and so the presence of an abnormally high gradient may indicate the formation of organized thrombus (known as pannus), impairing leaflet mobility and resulting in orifice obstruction. Elevated Doppler gradients may occur with a completely normal prosthesis because the phenomena of pressure recovery[40,43] increased the flow across the valve from a high-output state (anemia, fever, hyperthyroidism, large fistula) or if the surgical prosthesis is too small for the patient, that is, a "patient prosthesis mismatch."[40,44]

Aortic Balloon Valvuloplasty

Aortic balloon valvuloplasty is a catheter-based method to treat calcific aortic stenosis. From a femoral approach, the aortic valve is crossed and a stiff guidewire placed in the left ventricle. A balloon 16 to 24 mm in diameter and 3 to 5 cm long is briefly inflated across the aortic valve in an effort to widen the orifice. This technique reduces the obstruction from severe aortic stenosis to only a modest degree. Most patients will still have severe stenosis. Unlike mitral stenosis from commissural fusion, patients with calcific aortic stenosis have large collections of calcification within and around the valve, and there is little commissural fusion. Thus valvuloplasty has little effect on the underlying pathology. Typically, the average valve area increases from 0.5 to 0.8 cm^2, peak-to-peak systolic pressure gradients decrease from 55 to 29 mm Hg, and small improvements in LVEDP, pulmonary artery pressure, and cardiac output are noted.[45,46] Long-term outcome is poor with a very high restenosis rate, and the survival is similar to the natural history of untreated severe aortic stenosis.[47] In the current era, aortic balloon valvuloplasty is used primarily as a palliative method to treat calcific aortic stenosis in those with debilitating symptoms or refractory heart failure who are not candidates for valve replacement. It is sometimes used as a method to improve patients with acute decompensated heart failure, making them better candidates for surgical or TAVR. Fig. 6.26 is an example of the effect of balloon valvuloplasty on the transvalvular gradient in a patient with severe aortic stenosis and refractory heart failure, who was not a candidate for aortic valve replacement. A potential complication of aortic balloon valvuloplasty is the development

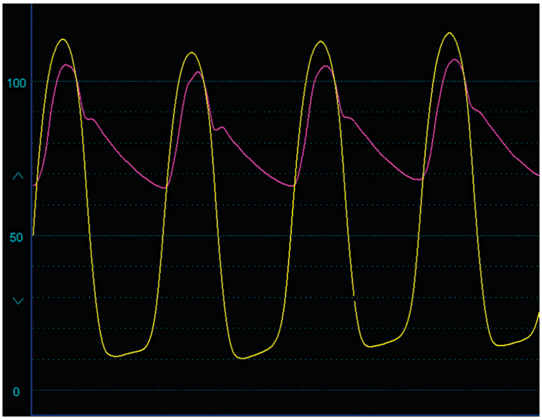


Fig. 6.25 Simultaneous left ventricular and central aortic pressures in a patient with a St. Jude aortic valve prosthesis (St. Jude Medical, St. Paul, Minnesota). The left ventricular pressure was measured with a 0.014-inch pressure wire with a micromanometer positioned into the left ventricle, using a retrograde approach.

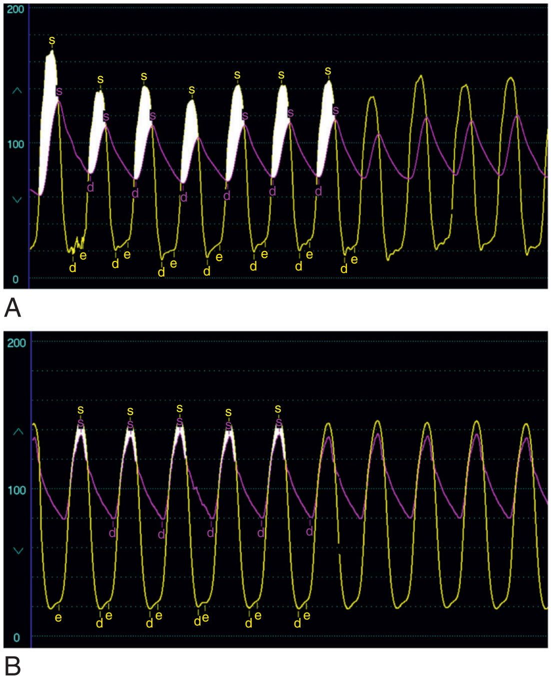


Fig. 6.26 Results of an aortic balloon valvuloplasty performed in an elderly male with severe aortic stenosis and severe refractory heart failure. This was performed as a palliative procedure as the patient was not an operative candidate. (A) The transvalvular gradient was approximately 35 mm Hg prevalvuloplasty. (B) The transvalvular gradient was reduced to less than 10 mm Hg after the procedure. *d*, Diastole; *e*, end diastole; *s*, systole.

of acute severe aortic regurgitation. This may be manifest by the development of elevated LVEDP and diastasis on the aortic waveform (Fig. 6.27).

Transcatheter Aortic Valve Replacement

One of the most dramatic and "game changing" medical breakthroughs in the field of interventional cardiology in the past decade is the emergence of transcatheter approaches to treat aortic stenosis. Prior to this technology, many patients with severe symptomatic aortic stenosis were untreated because they were considered high risk for open surgical replacement, the only option then available. TAVR, using balloon-expandable or self-expanding valve designs, was initially studied in this population of inoperable and high-risk patients and shown to be safe and effective with lower mortality than medical therapy, less morbidity than surgery, and improvement in quality of life.[48–53] This technology has been expanded to intermediate- and, more recently, low-risk patients and shown to be at least as good as surgery but with lower morbidity.[54–58] Based on these landmark clinical trials, TAVR has become the dominant method of treating aortic stenosis in all categories of patient risk.

Hemodynamic assessment is an important part of the TAVR procedure, particularly in the assessment of the post-TAVR results. Unlike surgically implanted valves with their sewing rings and struts, the profile of the balloon-expandable and self-expanding valves is minimal and unobtrusive. Thus there is usually minimal gradient observed by invasive methods after implantation. The presence of a significant gradient in the cardiac catheterization laboratory may indicate undersizing or underexpansion of the prosthesis and should be addressed at the time of the procedure. Assuming that the operator has checked for unbalanced transducers, bubbles, or kinks, there are also other explanations for higher than expected gradients, including the presence of a subvalvular gradient (Fig. 6.28) or, more rarely, a gradient in the aorta (Fig. 6.29).

Paravalvular aortic regurgitation is a well-described limitation and a unique complication of TAVR. Operators rely on intraprocedural echocardiography and aortic root angiography for its detection, but quantification can be challenging. The presence of trivial or mild (1+) aortic regurgitation due to small paravalvular leaks

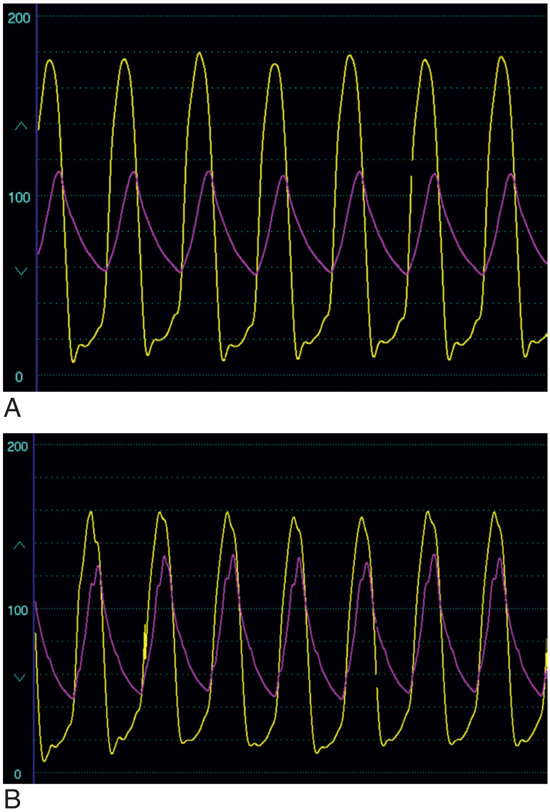


Fig. 6.27 Balloon aortic valvuloplasty may be complicated by aortic regurgitation. (A) The baseline hemodynamics from a 94-year-old male with refractory heart failure demonstrate a large transvalvular gradient from severe aortic stenosis. (B) Balloon valvuloplasty reduced the systolic gradient, but the aortic pulse pressure increased and the aortic diastolic pressure fell, consistent with the development of severe aortic regurgitation.

is not uncommon after TAVR. Moderate-to-severe paravalvular regurgitation carries a poor prognosis and is not acceptable postprocedure.[59–62] It is important to detect this during the procedure as additional intervention may be required to remedy it. Hemodynamic assessment post-TAVR may be helpful in this regard. A drop in aortic diastolic pressure with widening of the pulse pressure along with the elevation of the left ventricular diastolic pressure may be seen in the setting of severe paravalvular leaks (Fig. 6.30). However, the finding of a wide pulse pressure alone is a nonspecific finding and would not be expected in acute aortic regurgitation. One study found no relation between a low aortic diastolic pressure or a wide pulse pressure with the incidence or severity of aortic regurgitation.[63]

Several investigations explored various post-TAVR hemodynamic findings. Sinning et al.[64,65] described the aortic regurgitation index (AR index) to quantify the hemodynamic impact of a paravalvular leak (Fig. 6.31). The AR index is defined by the following formula:

$$\text{AR index} = \frac{\text{Diastoblic blood pressure} - \text{LVEDP}}{\text{Systolic blood pressure}}$$

The AR index relates to the severity of aortic regurgitation; the lower the index, the greater the degree of paravalvular regurgitation. An AR index of less than 25 was associated with poorer outcomes at 1 year.[64,65] Modifications and refinement of this index have been proposed by others.[66,67]

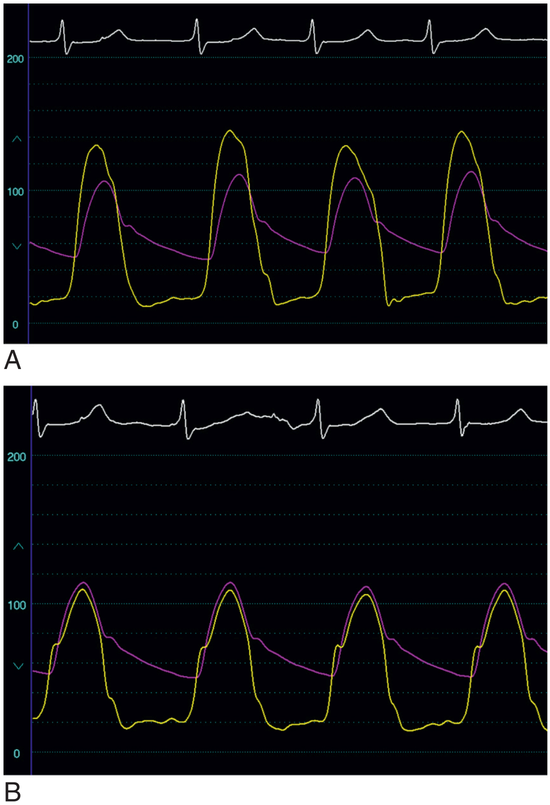


Fig. 6.28 An example of a subvalvular gradient in a patient undergoing a transcatheter aortic valve replacement (TAVR). (A) The hemodynamics post-TAVR measured with a catheter in the left ventricular apex reveal 25 to 30 mm Hg peak-to-peak gradient. (B) With careful pullback using an end-hole catheter, the gradient was determined to be subvalvular and not because of suboptimal valve position or sizing.

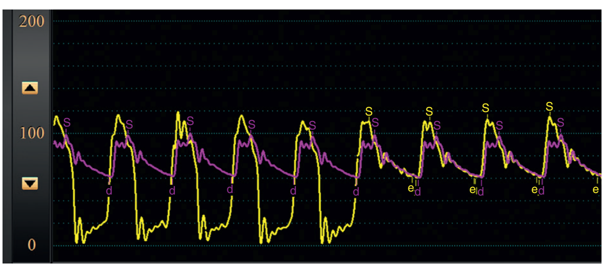


Fig. 6.29 Hemodynamics obtained from a 23-year-old male with a history of congenital aortic stenosis who had placement of an aortic homograft at age 9. He then developed severe aortic stenosis and underwent transcatheter aortic valve replacement. Note the presence of a gradient within the aorta during pullback of the catheter from the left ventricle to the aorta; this is due to stenosis of the homograft conduit. There is no gradient across the aortic valve. *d*, Diastole; *e*, end diastole; *s*, systole.

Paravalvular regurgitation during TAVR is often poorly tolerated. This is because the ventricle is compensated for stenosis and not regurgitation. The ventricular cavity is small and severely hypertrophied. Even small additional volumes will raise filling pressures and cause pulmonary edema or hypotension. Rapid pacing can be very useful to temporarily stabilize the decompensated patient with acute paravalvular regurgitation while efforts are being made to address this problem. This works because tachycardia shortens the diastolic filling period and reduces the degree of aortic regurgitation (Fig. 6.32).

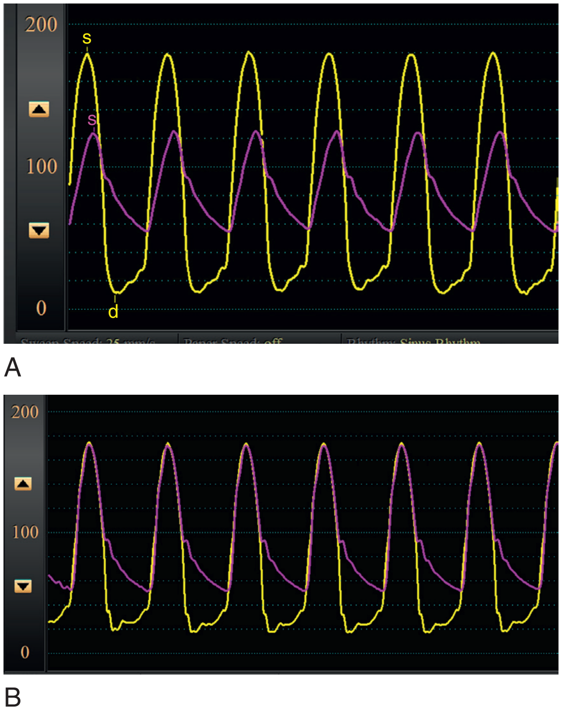


Fig. 6.30 Wide pulse pressure after transcatheter aortic valve replacement (TAVR) may indicate a paravalvular leak from aortic regurgitation. (A) Hemodynamics pre-TAVR show severe aortic stenosis. (B) Hemodynamics post-TAVR demonstrate no gradient but a large pulse pressure. *d*, Diastole; *s*, systole.

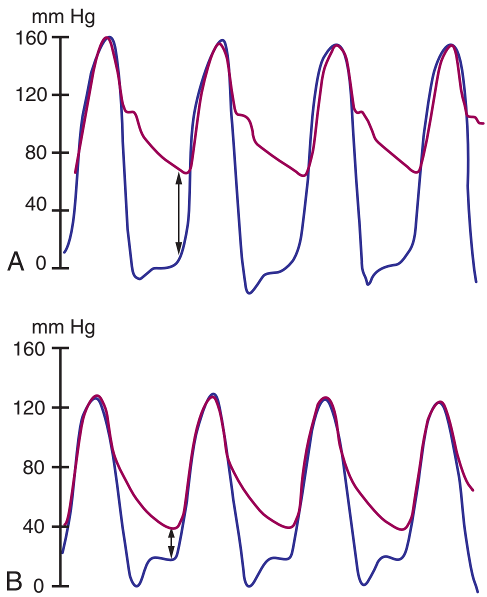
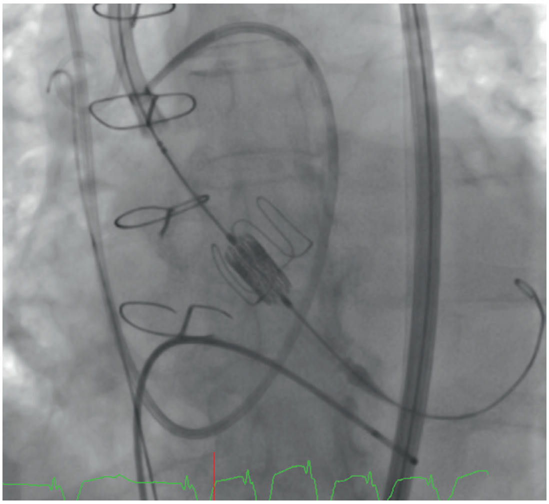


Fig. 6.31 (A) Simultaneous left ventricular end-diastolic pressure and diastolic aortic pressure in a patient without a paravalvular leak after transcatheter aortic valve replacement compared with (B) a patient with a paravalvular leak after transcatheter aortic valve replacement. The aortic regurgitation index for the first patient is 34.4 and for the second one is 15.4. The arrows show the difference between the diastolic blood pressure and the left ventricular end-diastolic pressure in the two scenarios. (From Sinning JM, Hammerstingl C, Vasa-Nicotera M, et al. Aortic regurgitation index defines severity of pre-prosthetic regurgitation and predicts outcome in patients after transcatheter aortic valve implantation. *J Am Coll Cardiol*. 2012;59:1134–1141.)

Fig. 6.32 The effect of heart rate on aortic regurgitation is shown. This patient underwent transcatheter aortic valve replacement and developed severe aortic regurgitation. On the left side of the panel, the patient is being paced at a heart rate of 95 beats per minute, the aortic pulse pressure is normal, and the left ventricular diastolic pressure is elevated at about 30 mm Hg. When pacing is stopped, the aortic pulse pressure widens, the left ventricular diastolic pressure increases, and there is diastasis, indicating severe aortic regurgitation.

A



B

Fig. 6.33 Example of transcatheter aortic valve replacement (TAVR) to treat prosthetic valve dysfunction. (A) There was severe aortic regurgitation in this 15-year-old bioprosthetic aortic valve with typical hemodynamic findings of diastasis and a small systolic gradient. (B) A balloon-expandable valve is centered on the bioprosthetic annulus and deployed with rapid ventricular pacing.

C

D

Fig. 6.33, cont'd (C) The hemodynamics post-TAVR reveal a rise in the aortic diastolic pressure, a fall in the left ventricular diastolic pressure, and resolution of diastasis. (D) Angiography post-TAVR revealed no regurgitation.

Transcatheter valve replacement is also used to treat patients with deteriorating bioprosthetic valves, the so-called "valve-in-valve" approach. An example of a "valve-in-valve" TAVR in a patient with a deteriorating bioprosthesis from central regurgitation is shown in Fig. 6.33. For patients with stenotic bioprosthetic valves, the decision regarding choice of TAVR versus open surgical repair should be made carefully as there may be substantial residual gradients present after TAVR, especially with smaller prostheses (Fig. 6.34).

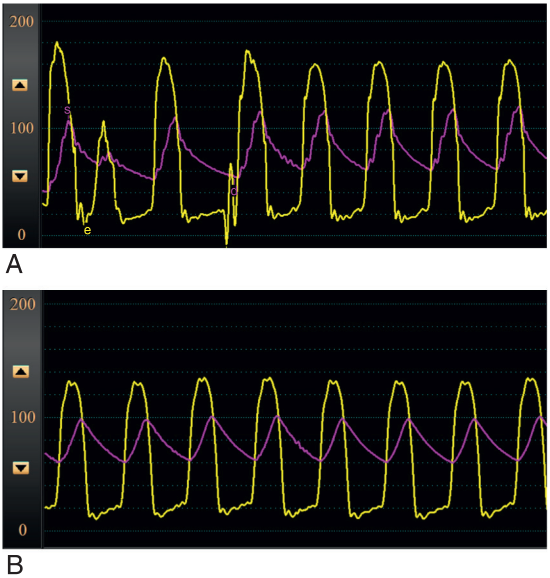


Fig. 6.34 The use of “valve-in-valve” transcatheter valve replacement (TAVR) for the treatment of a stenotic bioprosthetic aortic valve may result in substantial residual gradients, as shown in this patient undergoing TAVR with a self-expanding CoreValve (Medronic, Minneapolis, Minnesota) for the treatment of a stenotic, 21-mm porcine bioprosthesis. (A) Pre-TAVR. (B) Post-TAVR with a significant residual gradient present. *d*, Diastole; *e*, end diastole; *s*, systole.

References

1. Bekeredjian R, Grayburn PA. Valvular heart disease: aortic regurgitation. *Circulation*. 2005;112:125–134.
2. Carabello BA. Progress in mitral and aortic regurgitation. *Prog Cardiovasc Dis*. 2001;43:457–475.
3. Alexopoulos D, Sherman W. Unusual hemodynamic presentation of acute aortic regurgitation following percutaneous balloon valvuloplasty. *Am Heart J*. 1988;116:1622–1623.
4. Otto CM. Valvular aortic stenosis. Disease severity and timing of intervention. *J Am Coll Cardiol*. 2006;47:2141–2151.
5. Murgo JP. Systolic ejection murmurs in the era of modern cardiology. What do we really know? *J Am Coll Cardiol*. 1998;32:1596–1602.
6. Freeman RV, Otto CM. Spectrum of calcific aortic valve disease. Pathogenesis, disease progression, and treatment strategies. *Circulation*. 2005;111:3316–3326.
7. Park SJ, Enriquez-Sarano M, Change SA, et al. Hemodynamic patterns for symptomatic presentations of severe aortic stenosis. *JACC Cardiovasc Imaging*. 2013;6:137–146.
8. Laskey WK, Kussmaul WG. Subvalvular gradients in patients with valvular aortic stenosis. Prevalence, magnitude and physiologic importance. *Circulation*. 2001;104:1019–1022.
9. Laskey WK, Kussmaul WG. Pressure recovery in aortic valve stenosis. *Circulation*. 1994;89:116–121.
10. Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2021;143:e72–e227. https://doi.org/10.1161/CIR.0000000000000923.
11. Ring L, Shah BN, Bhattacharyya S, et al. Echocardiographic assessment of aortic stenosis: a practical guideline from the British Society of Echocardiography. *Echo Res Pract*. 2021;8:G19–G59.
12. Baumgartner H, Hung J, Bermejo J, et al. Recommendations on the echocardiographic assessment of aortic valve stenosis: a focused update from the European Association of Cardiovascular Imaging and the American Society of Echocardiography. *J Am Soc Echocardiogr*. 2017;30:372–392.
13. Abbas AE, Pibarot P. Hemodynamic characterization of aortic disease states. *Cathet Cardiovasc Interv*. 2019;93:1002–1023.
14. Yang CS, Marshall ES, Fanari Z, et al. Discrepancies between direct catheter and echocardiography-based values in aortic stenosis. *Cathet Cardiovasc Interv*. 2016;87:488–497.
15. Gorlin R, Gorlin SG. Hydraulic formula for calculation of the area of the stenotic mitral valve, other cardiac valves and central circulatory shunts. *Am Heart J*. 1951;41:1–29.
16. Hakki AH, Iskandrian AS, Bemis CE, et al. A simplified valve formula for the calculation of stenotic cardiac valve areas. *Circulation*. 1981;63:1050–1055.
17. Bermejo J, Rojo-Alvarez JL, Antoranz JC, et al. Estimation of the end of ejection in aortic stenosis. An unreported source of error in the invasive assessment of severity. *Circulation*. 2004;110:1114–1120.
18. Fusman B, Faxon D, Feldman T. Hemodynamic rounds: transvalvular pressure gradient measurement. *Catheter Cardiovasc Interv*. 2001;53:553–561.

19. Adele C, Vaitkus PT, Tischler MD. Evaluation of the significance of a transvalvular catheter on aortic valve gradient in aortic stenosis: a direct hemodynamic and Doppler echocardiographic study. *Am J Cardiol*. 1997;79:513–516.
20. Omran H, Schmidt H, Hackenbroch M, et al. Silent and apparent cerebral embolism after retrograde catheterisation of the aortic valve in valvular stenosis: a prospective randomised study. *Lancet*. 1994;361:1241–1246.
21. Meine TJ, Harrison JK. Should we cross the valve: the risk of retrograde catheterization of the left ventricle in patients with aortic stenosis. *Am Heart J*. 2004;148:41–42.
22. Hays J, Lujan M, Chilton R. Aortic stenosis catheterization revisited: a long sheath single puncture technique. *J Invasive Cardiol*. 2006;18:262–267.
23. Brogan WC, Lange RA, Hillis LD. Accuracy of various methods of measuring the transvalvular pressure gradient in aortic stenosis. *Am Heart J*. 1992;123:948–953.
24. Assey ME, Zile MR, Usher BW, et al. Effect of catheter positioning on the variability of measured gradient in aortic stenosis. *Cathet Cardiovasc Diagn*. 1993;30:287–292.
25. Carabello BA, Barry WH, Grossman W. Changes in arterial pressure during left heart pullback in patients with aortic stenosis: a sign of severe aortic stenosis. *Am J Cardiol*. 1979;44:424–427.
26. Schaefer S, Malloy CR, Schmitz JM, Dehmer GJ. Clinical and hemodynamic characteristics of patients with inducible pulsus alternans. *Am Heart J*. 1988;115:1251–1257.
27. Silver K, Aurigemma G, Krendel S, et al. Pulmonary artery hypertension in severe aortic stenosis: incidence and mechanism. *Am Heart J*. 1993;125:146–150.
28. Malouf JF, Enriquez-Sarano M, Pellika PA, et al. Severe pulmonary hypertension in patients with severe aortic stenosis: clinical profile and prognostic implications. *J Am Coll Cardiol*. 2002;40:789–795.
29. Yong G, Ali A, Feldman T. Diastolic transmitral valve pressure gradients in patients with severe calcific aortic stenosis. *Cathet Cardiovasc Interv*. 2009;74:957–964.
30. Carabello BA, Green LH, Grossman W, et al. Hemodynamic determinants of prognosis of aortic valve replacement in critical aortic stenosis and advanced congestive heart failure. *Circulation*. 1980;62:42–48.
31. Annabi MS, Touboul E, Dahou A, et al. Dobutamine stress echocardiography for management of low-flow, low-gradient aortic stenosis. *J Am Coll Cardiol*. 2018;71:475–485.
32. Nishimura RA, Grantham JA, Connolly HM, et al. Low-output, low-gradient aortic stenosis in patients with depressed left ventricular systolic function: the clinical utility of the dobutamine challenge in the catheterization laboratory. *Circulation*. 2002;106:809–813.
33. Monin JL, Quere JP, Monchi M, et al. Low-gradient aortic stenosis: operative risk stratification and predictors for long-term outcome: a multicenter study using dobutamine stress hemodynamics. *Circulation*. 2003;108:319–324.
34. Hachicha Z, Dumesnil JG, Bogaty P, Pibarot P. Paradoxical low flow, low gradient severe aortic stenosis despite preserved left ventricular ejection fraction is associated with higher afterload and reduced survival. *Circulation*. 2007;115:2856–2864.
35. Pibarot P, Dumesnil JG. Low-flow, low-gradient aortic stenosis with normal and depressed left ventricular ejection fraction. *J Am Coll Cardiol*. 2012;60:1845–1853.
36. Eleid MF, Nishimura RA, Sorajja P, Borlaug BA. Systemic hypertension in low-gradient severe aortic stenosis with preserved ejection fraction. *Circulation*. 2013;128:1349–1353.
37. Little SH, Chan KL, Burwash IG. Impact of blood pressure on the Doppler echocardiographic assessment of severity of aortic stenosis. *Heart*. 2007;93:848–855.
38. Lloyd JW, Nishimura RA, Borlaug BA, Eleid MF. Hemodynamic response to nitroprusside in patients with low-gradient severe aortic stenosis and preserved ejection fraction. *J Am Coll Cardiol*. 2017;70:1339–1348.
39. Cannon SR, Richards KL, Crawford MH, et al. Inadequacy of the Gorlin formula for predicting prosthetic valve area. *Am J Cardiol*. 1988;62:113–116.
40. Herrmann HC, Pibarot P, Wu C, et al. Bioprosthetic aortic valve hemodynamics: definitions, outcomes, and evidence gaps. *J Am Coll Cardiol*. 2022;80:527–544.
41. Kiefer TL, Wang A, Harrison JK. Invasive hemodynamic evaluation in patients with mechanical aortic valves. *Catheter Cardiovasc Interv*. 2013;82:43–50.
42. Michaels AD, Mester D. Ex vivo hemodynamic assessment of mechanical aortic valve gradients using a high-fidelity pressure wire. *Catheter Cardiovasc Interv*. 2011;77:726–732.
43. Eleid MF, Thomas JD, Nishimura RA. Increased prosthetic valve gradients: abnormal prosthetic valve function or pressure recovery? *Catheter Cardiovasc Interv*. 2014;84:908–911.
44. Daneshvar SA, Rahimtoola SH. Valve prosthesis-patient mismatch (VP-PM). A long term perspective. *J Am Coll Cardiol*. 2012;60:1123–1135.
45. NHLBI Balloon Valvuloplasty Registry Participants.. Percutaneous balloon aortic valvuloplasty. Acute and 30 day follow-up results in 674 patients from the NHLBI Balloon Valvuloplasty Registry. *Circulation*. 1991;84:2383–2397.
46. Otto CM, Mickel MC, Kennedy JW, et al. Three-year outcome after balloon aortic valvuloplasty. Insights into prognosis of valvular aortic stenosis. *Circulation*. 1994;89:642–650.
47. Lieberman EB, Bashore TM, Hermiller JB, et al. Balloon aortic valvuloplasty in adults: failure of procedure to improve long term-term survival. *J Am Coll Cardiol*. 1995;26:1522–1528.
48. Leon MB, Smith CR, Mack M, et al. Transcatheter aortic-valve implantation for aortic stenosis in patients who cannot undergo surgery. *N Engl J Med*. 2010;363:1597–1607.
49. Smith CR, Leon MB, Mack MJ, et al. Transcatheter versus surgical aortic valve replacement in high-risk patients. *N Eng J Med*. 2011;364:2187–2198.
50. Popma JJ, Adams DH, Reardon MJ, et al. Transcatheter aortic valve replacement using a self-expanding bioprosthesis in patients with severe aortic stenosis at extreme risk for surgery. *J Am Coll Cardiol*. 2014;63:1972–1981.
51. Kapadia SR, Leon MB, Makkar RR, et al. 5-Year outcomes of transcatheter aortic valve replacement compared with standard treatment for patients with inoperable aortic stenosis (PARTNER 1): a randomised controlled trial. *Lancet*. 2015;385:2485–2491.
52. Mack MJ, Leon MB, Smith CR, et al. 5-Year outcomes of transcatheter aortic valve replacement or surgical aortic valve replacement for high surgical risk patients with aortic stenosis (PARTNER 1): a randomised controlled trial. *Lancet*. 2015;385:2477–2484.
53. Gleason TG, Reardon MJ, Popma JJ, et al. 5-Year outcomes of self-expanding transcatheter versus surgical aortic valve replacement in high-risk patients. *J Am Coll Cardiol*. 2018;72:2687–2696.

54. Leon MB, Smith CR, Mack MJ, et al. Transcatheter or surgical aortic-valve replacement in intermediate-risk patients. *N Engl J Med*. 2016;374:1609–1620.
55. Thourani VH, Kodali S, Makkar RR, et al. Transcatheter aortic valve replacement versus surgical valve replacement in intermediate-risk patients: a propensity score analysis. *Lancet*. 2016;387:2218–2225.
56. Reardon MJ, Van Mieghem NM, Popma JJ, et al. Surgical or transcatheter aortic-valve replacement in intermediate-risk patients. *N Engl J Med*. 2017;376:1321–1331.
57. Leon MB, Mack MJ, Hahn RT, et al. Outcomes 2 years after transcatheter aortic valve replacement in patients at low surgical risk. *J Am Coll Cardiol*. 2021;77:1149–1161. 7.
58. Forrest JK, Deeb GM, Yakubov SJ, et al. 2-Year outcomes after transcatheter vs surgical aortic valve replacement in low risk patients. *J Am Coll Cardiol*. 2022;79(9):882–896.
59. Hayashida K, Lefevre T, Chevalier B, et al. Impact of post-procedural aortic regurgitation on mortality after transcatheter aortic valve implantation. *JACC Cardiovasc Interv*. 2012;5:1247–1256.
60. Genereux P, Head SJ, Hahn R, et al. Paravalvular leak after transcatheter aortic valve replacement. The new Achilles' heel? A comprehensive review of the literature. *J Am Coll Cardiol*. 2013;61:1125–1136.
61. Laakso T, Laine M, Moriyama N, et al. Impact of paravalvular regurgitation on the mid-term outcome after transcatheter and surgical aortic valve replacement. *Eur J Cardiothorac Surg*. 2020;58:1145–1152.
62. Pibarot P, Hahn RT, Weissman NJ, et al. Association of paravalvular regurgitation with 1-year outcomes after transcatheter aortic valve replacement with the SAPIEN 3 valve. *JAMA Cardiol*. 2017;2(11):1208–1216.
63. Parikh PJ, Romeiser JL, Dhautel B, et al. Predictors and impact of low diastolic blood pressure and widened pulse pressure following transcatheter aortic valve replacement. *Cardiovasc Revasc Med*. 2022;39:20–25.
64. Sinning JM, Hammerstingl C, Vasa-Nicotera M, et al. Aortic regurgitation index defines severity of pre-prosthetic regurgitation and predicts outcome in patients after transcatheter aortic valve implantation. *J Am Coll Cardiol*. 2012;59:1134–1141.
65. Sinning JM, Stundl A, Pingel S, et al. Pre-procedural hemodynamic status improves the discriminatory value of the aortic regurgitation index in patients undergoing transcatheter aortic valve replacement. *JACC Cardiovasc Interv*. 2016;9:700–711.
66. Mas-Piero S, Weiler H, Papadopoulos N, et al. Post/preprocedural ratio of hemodynamically assessed aortic regurgitation index as a marker for the need for corrective measures during transcatheter valve replacement: a first confirmatory study in patients receiving a new generation transcatheter self-expnadable prosthesis. *Catheter Cardiovasc Interv*. 2019;93:532–537.
67. Alkhouli M, Eleid MF, Nishimura RA, Rihal CS. The role of invasive hemodynamics in guiding contemporary transcatheter valvular interventions. *J Am Coll Cardiol Intv*. 2021;14:2531–2544.